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Perspective of Combustion Modeling for Gas Turbine Combustors

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ABSTRACT

A status summary report is given on the development of comprehensive computational combustor design (CCD) tool activities of GEAE combustion group conducted for the last three years. A "no-fudge factor" pre-test prediction approach is being pursued for the combustors that use rich or lean-dome swirl cups (Single Annular and Dual Annular combustors, SAC, DAC); Twin Annular Premixing Swirler, TAPS (SAC and DAC), and Dry Low Emissions' DACRS mixers for triple annular combustor domes. The pretest prediction capability is being gauged with existing data including emissions, exit temperature quality and wall temperatures from nine different combustors selected from a group of production and technology combustors. The agreement between predictions and data is good considering this is the 1st phase of the "no-fudge-factor" modeling approach. An integrated diffusion system/flame tube calculation approach is recommended for the swirl cup combustors. However for combustion systems with little dilution air, we can perform flame tube calculations with inlet conditions provided from through-the-inlets (e.g., vanes of the mixer) calculations. 4 to 10 million adaptive node calculations are envisioned for the both types of combustors before one could achieve a "grid independent" solution.

We are working closely with the advanced diagnostics teams of UC, UD/AFRL, NASA GRC and Sandia; and LES modelers from Georgia Tech, Penn State, Sandia and NRL to further improve pretest prediction capabilities for gas turbine combustor design.

INTRODUCTION

Gas turbine combustion modeling for supporting design effort was initiated in the early 1970's,that resulted in the first published successful demonstration of the use of the Computational Combustion Dynamics (CCD) tools in the combustor design process e.g., Mongia et al[1-4]. Several status and overview articles by Mongia and his coworkers have since appeared to share the continuing improvements in the combustion design tools and increased reliance on CCD for designing combustors; e.g. the most recent References 5 and 6. As noted in these publications, the design tools have progressively moved through the following three stages of improvements:

- Empirical/analytical design tools
- Hybrid modeling for analyzing current combustors
- Anchored design tools.

Table 1 and Figure 1 give a partial list of the gas turbine combustors that have used variants of empirical/ analytical design methodology proposed in Reference 1. The biggest deficiency of this methodology was that it provided qualitative guidance only. Due to many obvious reasons, including the number of grids one could afford, the highly simplified combustion models, and lack of benchmark quality data to propose and validate advanced models, Rizk and Mongia proposed a hybrid modeling approach. With a set of empirical constants, this approach gave an excellent agreement with 10 different combustors in regard to pattern factor (PF), smoke, lean blow-out fuel/air ratio, unburned hydrocarbons, CO and NOx emissions indices as shown typically in Figure 2. This clearly demonstrates potential for application of this modeling approach for designing new combustors. To-date there has not been reported however any application of the hybrid modeling for designing combustors.

The anchored design methodology[7] has addressed many of the limitations of the empirical/analytical design methodology. It does a reasonable job of simulating the complex combustor geometry with affordable number of nodes sometime exceeding one million. The state-of-the-art combustion models are employed including reasonable assumptions about the critical boundary conditions. Special emphasis has been placed on modeling the swirl cup and validating the analytical results with good quality data. The "same process and procedures" are used for different combustors so as to validate predictions with measured data on NOx, HC and CO emissions, and combustor exit profile and pattern factor, shown typically in Figure 3 for the selected five combustors. The anchored design methodology is used in all GEAE New combustion Product Introduction (NPI) programs; some of these are shown in Figure 4.

The anchored design methodology[7] has worked as well as originally intended. Its major deficiency continues to be its reliance on engine, rig and sub-component data for anchoring, and consequently its obvious limitation for achieving the desired pre-test prediction reliability. Its limitation became quite obvious during the design and development effort of the GE90 and CFM TAPS[8]. TAPS combustion system aero design and development process relied extensively on systematic use of rig (single module,

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sector and full-annular test rigs) and engine testing. The state-of-the-art CCD-based design tools[5-7] provided insignificant pre-test insight during the development phase of TAPS, Figure 5; however, they were useful for providing qualitative understanding of the test results.

"No-Fudge Factor" Design Tools

Therefore, the recent effort of the GEAE combustion design tools team is focused on developing the next-generation combustion design tools for achieving the following objective:

Develop a "pre-test prediction capability" that is as good as running the engine; which means model's accuracy and reliability are equivalent to engine data for accuracy and σ levels.

Specifically, this means achieving:

$\sigma_{TO\ NOxEI} < 0.69$ or 2.4% of take-off NOxEI; Idle $\sigma_{HCEI} < 1.25$, $\sigma_{COEI} < 4.4$; take-off and climb-out smoke number's $\sigma < 1.35$. Here σ denotes the root mean square error between model prediction and engine data. The subscripts identify the particular variable. Specifically $\sigma_{TO\ NOxEI} < 0.69$ or 2.4%; means that the take-off NOx emissions index (NOxEI) needs to be predicted within σ of 0.69 EI or 2.4% of the takeoff NOxEI. Similarly, idle $\sigma_{HCEI} < 1.25$, $\sigma_{COEI} < 4.4$ imply the σ values for idle HCEI and COEI should not exceed 1.25 and 4.4, respectively. The σ levels for liner/dome temperature, lean blowout (LBO) fuel-air ratio (FAR), combustor exit temperature profile and pattern factor (PF) should be 25°F, 0.0005, 0.01 and 0.03, respectively.

These highly aggressive goals for accuracy and pre-test predictive reliability (without supporting rig and engine testing) can be achieved by assuming that accurate combustion modeling methodology can be developed specific to particular types (classes) of gas turbine combustors. For example, this means that the GEAE combustion design tools team needs to develop models specifically for the following 5 types of combustors:

(a) Rich-dome swirl-cup based combustors for propulsion and industrial engines
(b) Lean-dome swirl-cup combustors
(c) Trapped Vortex Combustors (TVC)
(d) Twin Annular Premixing Swirlers (TAPS) combustors
(e) Dry Low Emissions (DLE) industrial engine combustors

Since the rich-dome swirl-cup combustors have been improved continuously during the last 30 years, we have an extensive empirical and fundamental understanding for this class of combustors. Moreover, this is a mostly diffusion flame combustors as contrasted with the mostly premix DLE combustors. Therefore, our initial effort for developing the design tools is focused on the rich-dome swirl-cup and DLE combustors. To-date, our focus has been on the nine combustors comprised of single annular, dual annular and triple domes shown schematically in Figure 6. The various fuel/air mixing devices under study include Dual Annular Counter-Rotating Swirlers (DACRS), two types of swirl cups (primary swirler stream created by discrete jets or radial inflow vanes), Triple Annular Research Swirls (TARS), and a partially premixing device, as shown by representative solid models in Figure 7.

The engine data to be used for tools validation include:

(a) Gaseous and smoke emissions;

(b) Dome and liner wall temperature levels;

(c) Pressure losses;

(d) Engine flameout, ignition (ground and altitude) data.

In addition, we use rig data for:

(a) Profile and pattern factor;

(b) Lean blowout, ignition and flame propagation;

(c) Pressure and air flow distribution around the combustor; and

(d) Pressure, temperature and FAR (fuel/air ratio) derivatives of CO, HC and NOx.

The CCD calculations must satisfy several requirements including the following four relevant to achieving CCD reliability for combustor design:

- The critical design features should be simulated to the accuracy levels consistent with design experience. This means, for example, the grid generation should capture details within the currently allowed dimensional and flow tolerances on product engine combustion systems, Figure 11.
- All the inlet conditions should be calculated by using through the inlets (vanes and holes of the swirl cups; primary and dilution holes; cooling holes, dome splash plate; refer to Figures 6 and 7) gridding and calculations. When possible, these calculations should be checked against measurements, Figures 8 through 10.
- The calculated airflow, pressure and loss distribution around the combustor should agree with measurements within the experimental σ levels, Table 2.
- CCD calculations should be grid independent. This means using adaptive grid generation technique and showing that the calculated results of interest to designer between the two successive levels of grid refinements stay within the engine data scatter bands. As a minimum, gridding morphology should be similar enough between the different combustors that

lack of agreement between predictions and data should not be attributed primarily to the normally mentioned CFD issues.

Unfortunately, none of the calculations reported in the literature on the practical devices satisfy all of the above requirements. Consequently, we indeed don't know how good or deficient the current or advanced models are as a predictive tool for engine combustors.

We have completed Phase I validation effort and have been working closely with the advanced diagnostics teams of UC, UD/AFRL, NASA GRC, Georgia Tech, Penn State and Sandia to conduct fundamental investigations on the main sub-components of the combustion system including fuel injector, swirl cup with and w/o dome splash plate, and primary/dilution air, and cooling air, Figures 6 and 7. These investigations are planned for a variety of combustor rigs including so-called single cup, flame tube and single module rigs. Other sub-component level investigations include liner cooling with or w/o the interacting swirl cup. These investigations need to conducted systematically to establish the model capabilities or deficiencies in simulating the critical processes relevant to gas turbine combustors including:

1. Complex fluid mechanics in the combustor diffusion system including pressure recovery, losses, flow separation and airflow distribution.

2. Turbulence and scalar transport modeling for swirling recirculating flows, jets in cross-flow, near-wall cooling films.

3. Liquid fuel injection, penetration, primary/secondary breakup, spray transport, heat-up/evaporation/burning.

4. Chemical kinetics, turbulence/chemistry interaction.

5. Soot formation and oxidation.

6. 3-D convective and radiation heat transfer in complex geometry with the participating non-uniform flowfield.

Integrated Combustion System Calculations

Integrated combustion system calculations can be performed with current commercially available codes, e.g., Fluent[9], as illustrated in Figure 12. It clearly shows the complexity of the nonreacting flowfield both within the diffusion system, inside the combustor and obvious interaction between the external and internal flowfield through the inlets. However, calculations of these types don't add much value in the combustor design process unless we establish the process and the associated accuracy levels achievable for the following:

1. Pressure and airflow distribution around the combustor.
2. Hydrodynamic validation of combustor liner subcomponents including mixer (fairly under control), primary/dilution holes, splash plate and cooling including nuggets and multi-hole.
3. Cold- and hot-side convective heat transfer.
4. Comprehensive injector/mixer modeling.
5. Turbulence/chemistry interaction including skeletal hydrocarbon and detailed NOx chemistry.
6. Soot formation/oxidation and radiation modeling.

Summary of our most recent work in this area is given in the following paragraphs.

Giridharan etal[18] have presented calculations for flow through dilution holes as a 1st step for dilution jet mixing model validation. They conducted a systematic study varying dilution diameter, wall thickness, pressure drop across the hole, Mach number and the boundary layer thickness of the inlet flow upstream of the dilution orifice. The primary emphasis of this study was to calculate grid-independent C_d and flow field characteristics at a plane of the hole along the wall on the hot side. The latter is for establishing jet inlet conditions for a typical gas turbine combustor, which has different values of cross-flow velocity component, u_{Hot} expressed in terms of J. Here J is the ratio of the dynamic head ($=\rho_{Hot}\ u_{Hot}^2/2$) to the liner pressure drop, ΔP. Grid independent solution was obtained for different liner pressure drops, K-factor (defined as $K=\Delta P_t/q_{annulus}$), J, hole diameters, wall thickness and the annulus boundary layer thickness.

They showed as an example, how the contours change for no cross-flow (J=0.0) and when the cross-flow dynamic head is 30% of the liner pressure drop (J=0.3). These calculations were done for a dilution hole diameter of 0.125 in. in a flat wall of 0.040 in. thickness; the liner pressure drop and K factor are 5.0% and 20, respectively. The calculated discharge coefficients are 0.6254 and 0.628. As expected, the predicted contours of the static pressure, the magnitude of the velocity vector and turbulent kinetic energy are significantly different for the two J-values. There is no easy way of calculating these types of values apriori, if one wants to separate calculations for the cold and hot-side of the combustor; hence the justification for an integrated diffuser/combustor calculations.

Diffuser Modeling

Karki etal[10] made a significant contribution to the art of combustor diffuser modeling even though there still remained several unresolved issues including turbulence modeling, grid resolution for 2D/3D CFD calculations. There is fairly good consensus on using the Shear-Stress Transport (SST) k-Ω model for flows involving adverse pressure gradients[11]. Our recent unpublished work[12,13] shows that one can't afford to map diffuser inlet profiles for a combustor diffuser rig (Figure 13a) fitted with bow-tie profilers (Figure 13b) to details and accuracy required

for diffuser model validation. Inlet profiles inferred from four 5-elemet rakes is not sufficient to resolve the flowfield characteristics. Consequently, 2-D CFD model results achieve mean square root error of ~13% (Table 2 Coumn A); an unacceptable inaccuracy level to conduct integrated combustion system analysis to support design effort.

However, it is now possible to calculate diffuser inlet conditions by using CFD based on fine-resolution 3-D grids starting upstream of the diffuser inlet, through the profiler and the prediffuser[12]; Figures 13c and d. This coupled with fine 2-D grid resolution of a diffusion system; Figure 13e provides a well-posed numerical solution approach[12]. Consequently, the resulting mean square root error is reduced to ~4%, Table 2 Column B. In a follow-up 3-D modeling study (with grids similar to Figure 13e), the discrepancy between model and data reduced[13] to ~2%, which is comparable to experimental error. What was interesting that the standard k-ε model was good enough to achieve the stated level of agreement with data from the diffuser, as is the case for the swirl cup modeling described in the following section.

Swirl cup & Dilution Data and Modeling

A unique fuel/air preparation device, called swirl cup, is used in several GEAE combustion systems, Figure 6-8, including CT7, TFE738, CF34, CFM56, CF6 and GE90. The critical design features of swirl cups have evolved over a number of years resulting in an excellent set of empirical design rules and tools, which have helped produce a series of low-emissions engines. In parallel with this has continued the development of physics-based design rules and tools.

We have recently initiated a comprehensive investigation to achieve the following objectives:

- An accurate phenomenological swirl cup model that can be used by designers during the conceptual design phase for conducting trade studies
- An accurate CCD-based swirlcup model that can be used during both the preliminary design and detail design phases without having to rely on the rig/analysis iterative process of the anchored design methodology[7].

Prof. San-Mou Jeng and his students have made benchmark quality nonreacting LV measurements of a typical swirl cup in a square box of 3″X3″ (76.2X76.2 mm) as described in Reference 14. The cup has the following dimensions considered important from a scaling point of view:

Flare exit diameter, D = 40 mm
Diameter of the venturi.= 17.5 mm
Outer and inner diameters of the secondary swirl passage are 27.2 and 21.6 mm, respectively.

Radial scans were taken at 53 axial stations each being 2 mm apart (from x = 3 mm to 107 mm), and additional 12 axial stations 4 mm apart for the region x = 111-155 mm. Here x is the axial distance downstream from the flare exit. For the first 16 axial stations (x = 3-33 mm), each radial scan included making spatial measurements of 73 radial locations (here we can denote them as y and z –direction). For the remaining axial stations the number of radial points were 37. The resulting data set for a fixed pressure drop of 4.0% includes mean and RMS values of u, v, w, and the two shear stresses uv, uw for a total of approximately 3000 individual spatial points. These results and preliminary observations are summarized by Mongia etal.[14]

In a follow-up study, Cai et al[15] have reported the effect of the test section size and hence the degree of confinement on the swirl cup flowfield by using six additional test sections of 4x4, 4.2x4.2, 4.3x4.3, 4.5x4.5, 5x5, 6x6 inches, respectively, in addition to unconfined flow. This effort has clearly shown that the flow field characteristics are distinctly different between strongly confined and mildly confined and unconfined flows. The flow behavior in the intermediate confinement exhibited bi-modal characteristics. Hsiao etal[16,17] have conducted parallel modeling activities to elucidate the fundamental processes occurring downstream of a typical swirl cup. The former has focused on solid modeling, parametric modeling, grid generation, and getting grid-independent solutions with the Reynolds averaged Navier-Stokes (RANS) equations. These activities have been very educational in regard to:

- Creating grids for the passages of the swirlers and simulating the details of the swirl cup, Figure 8.
- Splitting the calculation process into the two-steps – (1) doing through the vane calculations to establish the inlet conditions, and (2) swirl cup/test section flow field interaction.
- Use of tetra-, hexahedral or hybrid grids.
- Impact of inlet and outlet boundary conditions.
- Convergence and getting grid independent solution through a suitable adaptive methodology.

For example, the grid generation for the seven combustors investigated has established best practice for gas turbine combustors; shown for example in Figure 11 for a typical swirl cup combustor.

Flow field within the swirl cup and through the inlets is quite complicated, as shown in Figure 8. The first step toward achieving grid independence is getting a good agreement between the measured and calculated effective areas. The interpretation of the measured effective areas of

the individual passages may not be as straightforward as we thought before we started this phase of modeling activities. We achieved good agreement with the overall effective areas rather early in the process of grid-refinement. However, one to two additional grid refinement steps needed to be taken before "claiming grid-independence" for the complex 3-D flow field characteristics. We have not yet fully resolved what is the best approach for achieving grid independence. CFD has been very successful in reproducing the measured values of effective areas of several designs of all the five mixers shown previously in Figure 7. Figures 9 and 10 show typical comparison between data and predictions.

FLUENT[9] has been used by Giridharan etal[18] to calculate flow through dilution orifices for the range of pressure ratio parameter ($K= \Delta P_t/q_{ann}$) of interest in combustor design. Here ΔP_t and q_{ann} denote total pressure drop across the dilution hole, and dynamic head of the annular stream upstream of the dilution hole, respectively. The predicted discharge coefficients agree with data quite well; and jet angles look reasonable as shown in Figure 14.

Other main reason for the investigation of Giridharan et al[18] was to establish a process for defining inlet conditions for the primary and dilution jets if one were to do flame tube calculations separately instead of doing an integrated combustor flame tube and diffuser analysis. The latter can be done in principle. But there are several challenges and issues involved if one were to take this approach, including the following:

- The enormous effort involved in the grid generation process (Figures 11c and d)
- The complexity of the flow field and mutual dependence of the inlet conditions and the combustor internal flow field, as shown in Figure 12.

Grid Generation

There are many ways of creating grids for combustor applications. The two techniques for a typical combustor, Figure 4, tried by Hsiao and Vij[19] are explained briefly in the following paragraphs.

The first approach uses the hexahedral grid generation process used by ICEM which is based on the principle that a geometry can be approximated by cutting a large brick into several small ones so that all surface details are represented by external faces of the collection of smaller bricks. The set of all bricks put together is called a multi block topology. The edges and faces of each block in the topology must be projected manually to the corresponding faces and edges in the geometry. Next each edge must be assigned an edge count to generate the grid. The grid quality is measured by the determinant, which represents the closeness of the cell shape to a cuboid. The determinant of a perfect cuboidal cell would be 1.

The following strategy was used for creating grids for a one-nozzle sector of a typical combustion system:

1. Create a blocking for a single jet and a single vane of the swirl cup.
2. Copy and merge the jets and vanes at the multi block topology level.
3. Create unstructured grids for the jets and vanes.
4. Create a separate blocking for the swirl cup region surrounding the jets and vanes.
5. Create an unstructured grid from step 4 and merge with grids from step 3. This gives the full swirl cup grid.
6. Create the fuel nozzle stem grid by growing o-grid blocks by extrusion from the diffuser swirl cup interface.
7. Create a separate blocking file for the rest of the diffuser dome passage excluding the fuel nozzle.
8. Merge the blocking from steps 7 and 8. Create a full diffuser grid from the merged blocks.
9. Merge the unstructured grids from steps 5 and 9 to obtain the diffuser + swirl cup grid; e.g., 2.2 million cells shown in Fig. 11c.
10. Create a separate blocking for the combustor and create an unstructured grid for the combustor typically about one million cells as presented in Figure 11d.
11. Merge the unstructured grids from steps 9 and 10 to get the combined combustor + diffuser + swirl cup grid.

The diffuser + swirl cup grid was created in about 640 hr. and the combustor grid in about 200 hr. Given familiarity with the geometry and technique, a similar grid creation for future applications would would take considerably less effort to create. This estimate does not include time taken for aero-volume creation and geometry cleanup. The total grid size for the combustion system shown in Figures 11 is ~3.2 million cells. The determinant was > 0.1 and improved to > 0.3 after smoothing in ICEM. This grid covered volume within 0.03% and surface area within 0.7% for all tagged surfaces, compared to the actual geometry.

The 2nd grid generation approach used Gambit of FLUENT. A tetrahedral mesh of 2.4 million cells was created in Gambit's automated tetra mesher. It includes finer details such as the nugget holes and splash plate cooling holes. These features are approximated as quad faces in the ICEM hexa mesh. The tetra meshing effort included 120 hours for aero-volume creation including cleanup of tiny regions and 180 hours for creating and fine-tuning the grid in Gambit. The tetra mesh captures the volume to within 0.13% and surface area within 0.04%. The tetra grids quality is measured by skew and

aspect ratio, which were less than 0.89 and 6.7, respectively. It is estimated that similar future applications of Gambit automated tetra mesher would also take considerably less effort.

Even if one automates the grid generation process for the combustor and diffuser, we feel that the major drawback would still be with the inadequacy of predicting pressure distribution around the combustor. We expect to get a better agreement with data from gas turbine combustor diffuser in the near future.[12,13] How far this approach will be successful for multi-passage diffusers, Figure 6 is rather a speculation at this point in time. Nevertheless, an example showing why separate calculations for the diffusion system and the combustor can not be regarded as "no fudge factor pre-test prediction" is illustrated by focusing on the complexity of specifying the boundary conditions for a typical dilution hole, Figure 12.

Combustion models

The 1st phase of the turbulent combustion modeling, which started in 1997, focused on the flames relevant to nearly premix DACRS flames. The most recent laminar flamelet modeling results are summarized by Held etal[20] The laminar flamelet modeling effort driven by data covered a broad range of engine operation that included close to flameout conditions where one encounters a sudden increase in unburned hydrocarbons and CO. As shown in Figures 15, the predicted contours of CO and NO are significantly different at low and high flame temperatures of 1750 and 1900K, respectively. The qualitative comparison between predictions and data for NOx, CO and unburned HC is quite good as shown in Figure 15.

The DLE combustion systems modeling effort continues including the significant challenges because:

(a) The flames are operating slightly above LBO fuel-air ratios;
(b) The measured flowfield has turbulent flow structure which is more suitably modeled by LES;
(c) The sensitivity to the inlet conditions is more than the conventional diffusion flame combustors.

Held etal[20] started their calculations immediately downstream of the mixer exit plane with profiles estimated from a limited database. Recent study by Giridharan[22] for the DACRS mixer through the vanes calculations show that of the various turbulence models studied (RLZ, STD, RNG and RSM are calculations using the realizable k-ε, the standard k-ε, Renormalized k-ε, and the Reynolds stress models coded in FLUENT[9]), the realizable k-ε model gave the best agreement with data. For the swirl cup through the vane analysis by Hsiao etal[16,17] as presented in Figures 9 and 10 show:

- Current practice[7] based on 2-D axisymmetric calculations within the swirl cup with plug flow approximation don't agree well with data.
- Standard k-ε model gives best overall agreement with the extensive database created for the swirl cup under study.
- More work is needed to get good agreement with the recovery region.
- The measured flowfield is not as well ordered as RANS calculations show indicating the desirability of undertaking LES simulation.

Held etal[20] laminar flamelet model combined with through the inlets (including vanes, purge holes and the natural gas injector) calculation process for the swirl cup has achieved good agreement with an industrial engine emissions data for dry and wet-control NOx. This work by Giridharan etal[22] identified as combustor #3 (C3Dry and C3Wet in Figure 16) shows promise for the potential application of the Held etal laminar flamelet for both the DLE and diffusion flame combustors. Here, Comb2 and Comb4 denote DLE combustors. Model improvement effort work will continue to get simultaneous good agreement for Nox, CO and HC for industrial engine combustors with Held etal LFM approach, Dai et al[23].

An alternate laminar flamelet model proposed by Li etal[21] when applied to the industrial engine combustor studied by Giridharan etal[22]gives an excellent agreement with the same engine emissions database, as presented typically in Figure 17. This partially-premix opposed diffusion-flame model coupled with comprehensive injector modeling has been extended to several propulsion engine combustors.

Several approaches are currently being pursued for improving the use of the laminar flamelet modeling to flows involving essentially diffusion flame on one side and fully premixed on the other side. In addition, work continues for improving reduced-kinetic and skeletal mechanisms including even soot formation and oxidation. Our results from these studies will be coming out in future publications.

Reacting Spray Combustion models

Most of the prior gas turbine combustor modeling activities[1-7] has made arbitrary assumptions about the quality of spray. Several investigators have proposed comprehensive injector modeling approaches typical to Rizk etal[24-26] however, their application as an integral part of CCD is not well established. Recognizing the importance of spray quality in swirl cup and even more so for next-generation low-emissions combustors[8], we have

undertaken a rather extensive effort in this area during the last three years, as depicted schematically in Figure 18. Our preliminary calculations as presented in this section indicate the practical viability of this approach as demonstrated by application to the several of the combustors covered in this paper. Figure 18 lists the steps involved in predicting droplet size and velocity distribution from a typical air-atomizing device including swirl cup, simplex and prefilming air blast nozzles.

Figure 19a shows typical results for a round liquid jet injected into a cross-stream. The predicted behavior (penetration and droplet size) is in reasonable agreement with the limited available data. A similar approach has been used for modeling swirl cup filming, primary/secondary atomization, spray transport and evaporation., as shown in Figure 19b. Further model validation effort will continue consistent with the availability of the benchmark quality data relevant to GEAE engine and advanced combustors.

Application of Hsiao etal[27] comprehensive injector modeling, along with through the vane analysis[16,17] and laminar flamelet model of Li etal[21], give reasonable droplet size distribution for both the SAC and DAC combustors for the four typical operating points in the low-power region (Figure 20) where spray quality plays more significant impact than high-power operation, internal thermal field characteristics at high-power and comparison of combustor exit profile. The agreement between data and model predictions is rather quite encouraging, Figure 21, considering that five different combustors have been analyzed using the same approach.

Effect of Grids

In order to keep CCD turn-around time over-night without requiring massive computer resources, it has been a common practice of the GEAE combustion group to limit the grid size between 150-300K. Because of several reasons including the following, most of the numerical solutions for the diffusion system and the combustor internal flow field have been done separately.

(a) CFD with 100-200K nodes and the standard k-ε model gives only a qualitatively good agreement with data in regard to diffusion system losses, pressure and air flow distribution. However, with fine grids and advances in turbulence modeling[11] we can now have better agreement with data. Therefore, the 1st hurdle over getting the integrated combustion system analysis is almost behind us.

(b) The conventional convective and radiation heat transfer models are not good enough to make accurate predictions for wall temperature levels and gradient. However, with recent advances in this area[6], it is the right time to define what needs to be done to achieve integrated cold/hot side calculations.

(c) The current turbulent combustion models are only qualitatively accurate, and thereby make it very hard to predict liner and dome wall temperatures. However because of the recent emphasis on getting better quality database and a systematic evaluation of combustion models, industry needs to put in place bigger computer resources for undertaking large scale CCD projects for design.

From a designer's point of view, the grid independence solution implies that the variables affecting performance, emissions, operability and durability don't change significantly. If we assume that the main sources contributing to σ between data and predictions are numerical diffusion, calculated inlet conditions, turbulence, spray, fuel/air mixing, chemical kinetics, turbulent combustion, heat transfer, and others, and they are weighted equally, then achieving the σ levels of Table 3 implies getting a grid independent solution. It is well recognized that grid independence does not simply imply doubling the number of grids in the three spatial directions and showing no significant change in the flow field characteristics. One instead needs to keep on adding more grids where the "gradients" are the steepest until we get to the minimal acceptable level of numerical diffusion.

Table 3: Average σ and maximum σ (assumed 3 times of average σ) attributed to any one of the nine sources of errors*

Variable	Average σ	Maximum σ
Takeoff/climb NOxEI	0.23	0.69
Idle HCEI	0.42	1.25
Idle COEI	1.5	4.4
Takeoff/climb SAE Smoke #	0.45	1.35
Liner/dome temperatures, F	8	25
LBO FAR	0.0002	0.0005
Profile Factor	0.0033	0.01
Pattern Factor	0.01	0.03

*The sources of errors are:
Numerical diffusion; 2. Inlet conditions; 3. Turbulence models; 4. Comprehensive injection, spray transport, heat-up and evaporation models; 5. Fuel/air mixing models; 6. Chemical kinetic models; 7.Turbulent combustion models; 8. Heat transfer models; and 9. Others.

GEAE combustion started the effort on achieving grid independence for the nine combustors that include conventional, partially mixed and almost perfectly mixed combustors. As part of these studies, we need to investigate how adapting grids change the flow field, the flame structure, near-wall gas temperatures, heat transfer coefficients, NOx, CO, HC and LBO FAR. We have run

several steps of grid refinements starting with approximately 300K nodes and stepping up to 8 million nodes; the latter still is waiting to be processed. It appears that the solution of the 4-million case is still not grid independent. If we look at the adiabatic wall temperatures calculated with 300K and 4 million nodes, the numerical diffusion σ is 74°F. This is an order of magnitude higher than the goal of 8°F.

Integrated Cold/hot side calculations

In order to assess how far we can go without having to solve simultaneously the flow field external and internal to the combustor dome and liner (Figure 12), we have undertaken systematic evaluation for each of the following combustor subcomponents:

(a) Outer and inner combustor annuli with and w/o taking into account details of the cooling, primary and dilution holes.
(b) Dome cowling with and w/o flow through the swirl cup, splash plate and flare.
(c) Diffusion system with and w/o details of the fuel nozzle.

These investigations have clearly shown that we need to have 3-D integrated combustor/diffusion system analysis approach that includes details of the combustion system including diffuser, cowl, dome, combustor outer and inner liners, combustor outer and inner flow passages, and fuel nozzles, as shown schematically in Figure 12.

Joint Scalar PDF Modeling

The objective of this work[28] is to study the feasibility of transport joint scalar probability density function (JSPDF), in particular turn around speed, for modeling swirler cup to assist design optimization. Specifically, JSPDF has been applied in modeling the performance of a Dry Low-Emissions (DLE) lean premixer for marine and industrial engine application. Test data including emissions have been obtained for this premixer from single cup rig (SCR). Calculations using JSPDF model are therefore compared to these data.

The JSPDF $f(\phi_k)$ transport equation is written as

$$\frac{\partial}{\partial t}(\rho f)+\frac{\partial}{\partial x_i}(\rho u_i f)+\frac{\partial}{\partial \psi_k}(\rho S_k f)=-\frac{\partial}{\partial x_i}\left[\rho\left\langle u_i''|\psi\right\rangle f\right]+\frac{\partial}{\partial x_i}\left[\rho\left\langle\frac{1}{\rho}\frac{\partial J_{i,k}}{\partial x_i}\middle|\psi\right\rangle f\right]$$

The turbulent scalar flux term is modeled by the gradient-diffusion assumption:

$$-\frac{\partial}{\partial x_i}\left[\rho\left\langle u_i''|\psi\right\rangle f\right]=\frac{\partial}{\partial x_i}\left[\frac{\mu_t}{\rho \mathrm{Sc}_t}\frac{\partial f}{\partial x_i}\right]$$

The mixing term (second term on right hand) is modeled by the modified Curl's model, mixing constant = 2.0. Note that the chemical reaction (third term on left) is resolved in exact form within JSPDF approach. Monte-Carlo method is then adopted to solve this transport equation. The coupling of JSPDF with flow field is through turbulent viscosity (to JSPDF from flow field) and density (to flow field from composition by JSPDF). The flow field is modeled by the standard k-ε model. A simple four step reduced chemistry mechanism for methane, which is being developed by Li[29], was adopted in this study along with In-Situ Adaptive Tabulation (ISAT) method. The current calculations use the commercial CFD code FLUENT@ version 6.1 which implements the above models.

The studied premixer consists of pilot and main swirlers. The pilot is counter rotating swirler with 8 fuel injectors near the exit to enhance lean blow out (ELBO) capability. A 3D model of 45 degree SCR sector starting from main swirler exit and pilot throat is modeled with one ELBO fuel injector resolved. Total meshes is ~0.55 millions with about 20 particles each cell. Error tolerance is 1.0E-7 for ODE solver and 1.0E-5 for ISAT for the calculation of chemical reaction term. Velocity boundary conditions are defined from LDV measurement. Due to the lack of good measurement of fuel/air profiles, uniform profiles are specified for compositions at both inlets (i.e., at the pilot throat and main exit).

Conditions of two cases, and results of emissions are summarized in Table of Figure 22. All figures shown are from case 1. The axial velocity contour indicates the stagnation point is ~1 diameter downstream of pilot exit. Due to ELBO fuel at the pilot exit, localized high temperature is seen in the center-body recirculation zone, which contributes to a significant portion of NOx generation. Comparison to experiment data is shown in Table. The emission trend with flame temperature is captured, but the calculated NOx is lower than the measurement and CO has mixed indication. However, note in case 1, the flame is close to incipient blow-out indicated by CO, and thus CO measurement is not reliable. It is expected that better agreement can be obtained with more advanced chemistry mechanism (such as augmented reduced mechanism or ARM), and better input on fuel/air profiles for boundary conditions.

Computations are conducted on one single IBM PC workstation with two Intel processors of 2.6GHz. The maximum ISAT table size is set at 200MB. JSPDF calculations are initiated at eddy-breakup solutions. It generally takes more than 1 week clock time to complete one case while the machine is being used for other tasks at the same time. ISAT performance is shown in Figure 22. The average CPU time per iteration without reaction is about 1.8 minutes. At beginning of ISAT, the table is being

built up, and most of CPU time is taken to integrate the ODE for reaction rate. It shows that about 10 out 13 minutes or 70% of time are spent for chemical reaction calculation. By 2000 iterations, the total CPU time per iteration is reduced to about 3.3 minutes, i.e., the time to evaluate reaction rate is down to 1.5 minutes. Therefore ISAT has achieved about 7 times speed-up in computing reaction rate. Without ISAT algorithm, it is not feasible to achieve this 3D simulation of JSPDF (even with simple 4 step reaction mechanism) in a realistic time frame.

Although current results have not shown a satisfactory comparison of emission predictions to experiments. Further improvement of computer resources, such as parallel capability, can greatly enlighten the JSPDF ability to handle more detailed yet complicated chemistry mechanism along with ISAT, and thus achieve the goal of assisting real designs of low emission premixers. For example, one case of these calculations can be completed within days if multiple processors are to be used.

Large Eddy Simulation

There are many obvious reasons for using Large Eddy Simulation, LES for gas turbine combustors[29-33]. We have been working closely with several research groups involved in LES. For example, we follow closely Menon's work[29] since it focuses on dynamical structure and the relating modeling issues that impact gas turbine combustion processes for gas turbine combustor designers. It relies on RANS solution to provide inlet boundary conditions, and will adopt complex geometry simulation technology from our experience in gas turbine combustor RANS calculations. We have been working closely with Grinstein and his coworkers[30,31] since their emphasis is on the specification of the inlet conditions for LES. We are working closely with Professor Yang because the main thrust of his LES work[32] is characterization of combustion dynamics of gas turbine mixers that invariably have complex geometrical features. Finally the details of the near-wall regions need to be properly simulated with LES along with spray, which we rely on the research activities of Oefelein and Schefer[33].

These joint activities between GEAE, Georgia Tech, Penn Stat, NRL and Sandia have put the proper emphasis on addressing both fundamental and design driven applied research of particular interest to gas turbine combustor design community[34-36]. Specifically it means resolution of complex geometry, proper specification of the inlet/boundary conditions, NOx, CO and HC prediction. As reported in these papers, there is back-to-back comparison between RANS and LES for

- Swirl cup nonreacting flow field
- DLE reacting flow.

As mentioned by Shih and Liu[37], LES work needs to compliment RANS calculations. We find LES modeling effort as typically represented by Menon, Yang, Oefellein and their colleagues35-37 bringing us closer to applying LES techniques to designing next-generation gas turbine combustors for improving performance, operability and/or emissions.

Summary

A status summary is given in this paper of the last three years activities of GEAE combustion group for further improving the reliability and accuracy of predictive CCD tools for gas turbine combustor design. The prime objective of the on-going effort is:

> Develop a "pre-test prediction capability" that is as good as running the engine; which means model's accuracy and reliability are equivalent to engine data for accuracy and σ levels.

The desired root-mean-square (σ) level for the variables of interest is:

Variable	Desired σ
Takeoff/climb NOxEI	0.69
Idle HCEI	1.25
Idle COEI	4.4
Takeoff/climb SAE Smoke #	1.35
Liner/dome temperatures, F	25
LBO FAR	0.0005
Profile Factor	0.01
Pattern Factor	0.03

An integrated diffusion system/flame tube calculation approach is recommended for conventional diffusion flame combustors that employ primary and dilution holes. This is coupled with through the inlet calculations for the mixer, splash plate and cooling holes to provide inlet conditions consistent with the best models employed.

However for the DLE combustion systems or others where most of the combustion air enters through the dome, which rarely use primary or dilution holes, one can split the calculations into two separate sets, namely diffusion system, and the flame tube.

In order to claim "grid independence" it is envisioned that combustion systems would need 4-10 million adaptive grids before we can start sorting through the deficiencies of the different sub-models of relevance to production gas turbine combustors.

LES simulation techniques have made significant progress during the last five years, and are ready for helping the designers to design next-generation gas turbine combustors for enhancing performance, operability and/or emissions.

Acknowledgment

The author would like to express his thanks and gratitude to the Advanced Combustion CCD team members for contributing calculations and sharing their pre-published works in this paper.

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Table 1: Partial list of combustors designed with the empirical/analytical design methodology.

1. AF/Navy Trapped Vortex Combustors
2. CF6-80C Low Emissions Combustor (LEC) – **Anchored methodology**
3. GE90 Dual Annular Combustor (DACII) - **Anchored methodology**
4. Army/NASA Advanced Fuel Injector.
5. Army Low Pattern Factor Combustor.
6. NASA/Army Compliant Matrix Cooling Comb
7. 501K 9 PPM NOx Combustor
8. 501K 25 PPM NOx Combustor
9. NASA High Speed Ultralow NOx Combustion Conceptual Design Feasibility
10. DOE Advanced Coal-Fueled Turbine
11. DOE ATTAP (Automotive) Combustor
12. 501K Effusion Combustor.
13. 501K 80 PPM NOx Combustor.
14. T406/GMA 2100 Combustor.
15. GMA 3007 Combustor.
16. T56-A427 Nozzle Carboning.
17. Model 150 combustor
18. Model 120 Combustor.
19. T800+ Lamilloy Combustor.
20. T800+ Effusion-Cooled Combustor.
21. Low Smoke 570 combustor.
22. Expendable Turbine Engine Combustor
23. Limited Life Turbine Combustor
24. ATEGG XTC 16/1
25. TFE1042 combustor
26. F109 combustor.
27. Garrett's first ATEGG Combustor.
28. Advanced Cruise Missile Engine Combustor.
29. Low Smoke TFE 731
30. ATF-3 combustor
31. NASA Counter-Flow Film-Cooled Combustor.
32. DOE/NASA Automotive Gas Turbine Combustion Concept.
33. AF Variable Geometry Combustor.
34. AF Small Turbine Afterburner
35. AF Segmented Ceramic Combustor.
36. Navy Improved Combustor Dome.
37. Army Concept I increased to 2200^0F temperature rise.
38. DARPA/Navy Ceramic Combustor
39. ETF 131 Combustor and Afterburner.
40. NASA Pollution Reduction Technology (PRTP) Fuel Staged Concept 3.
41. NASA PRTP PM/PV Concept 3.
42. Army Combustor Design Criteria Concept I
43. Army Combustor Design Criteria Concept II

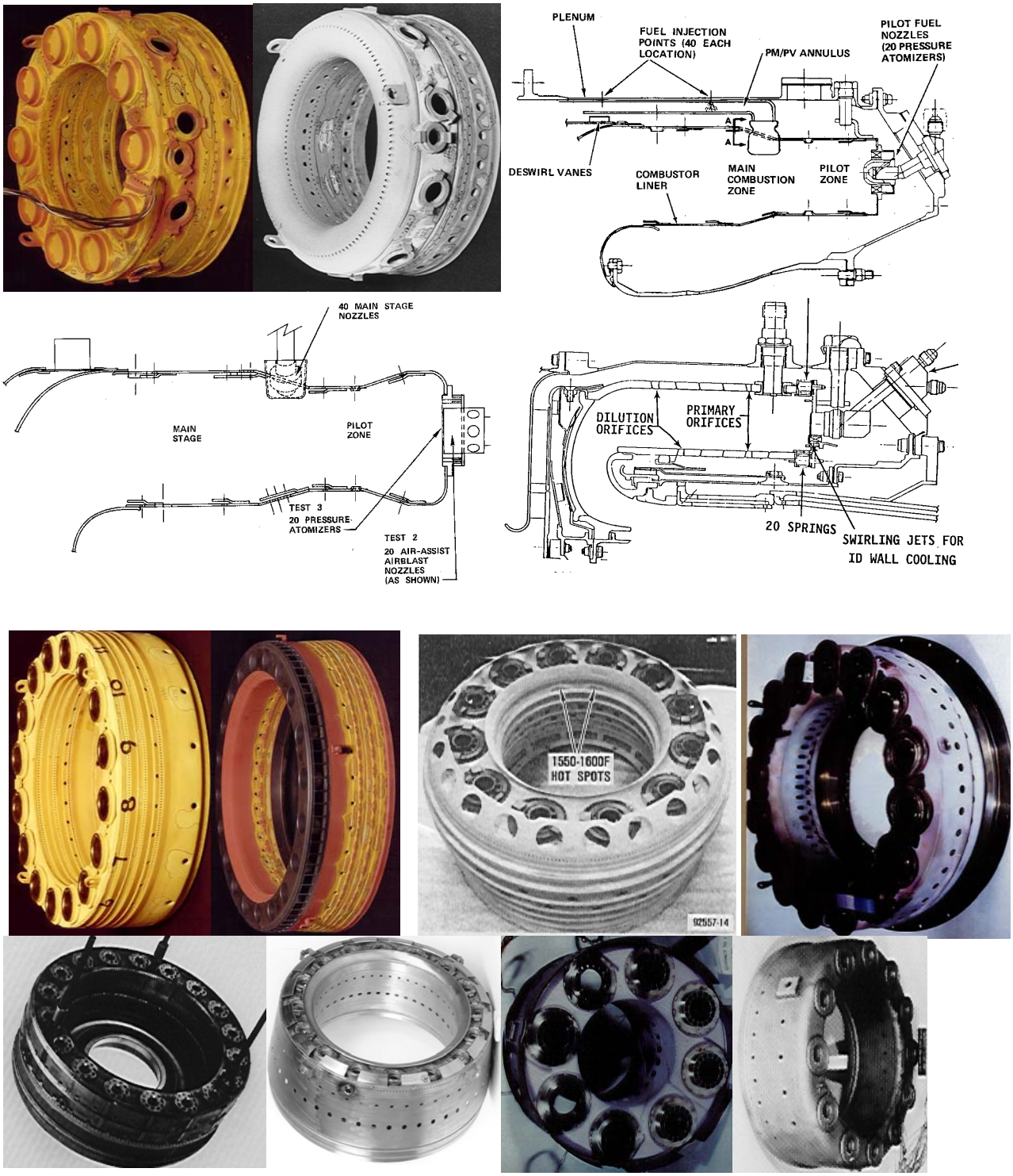


Figure 1: 13 out of 43 combustors designed with empirical/analytical design methodology.

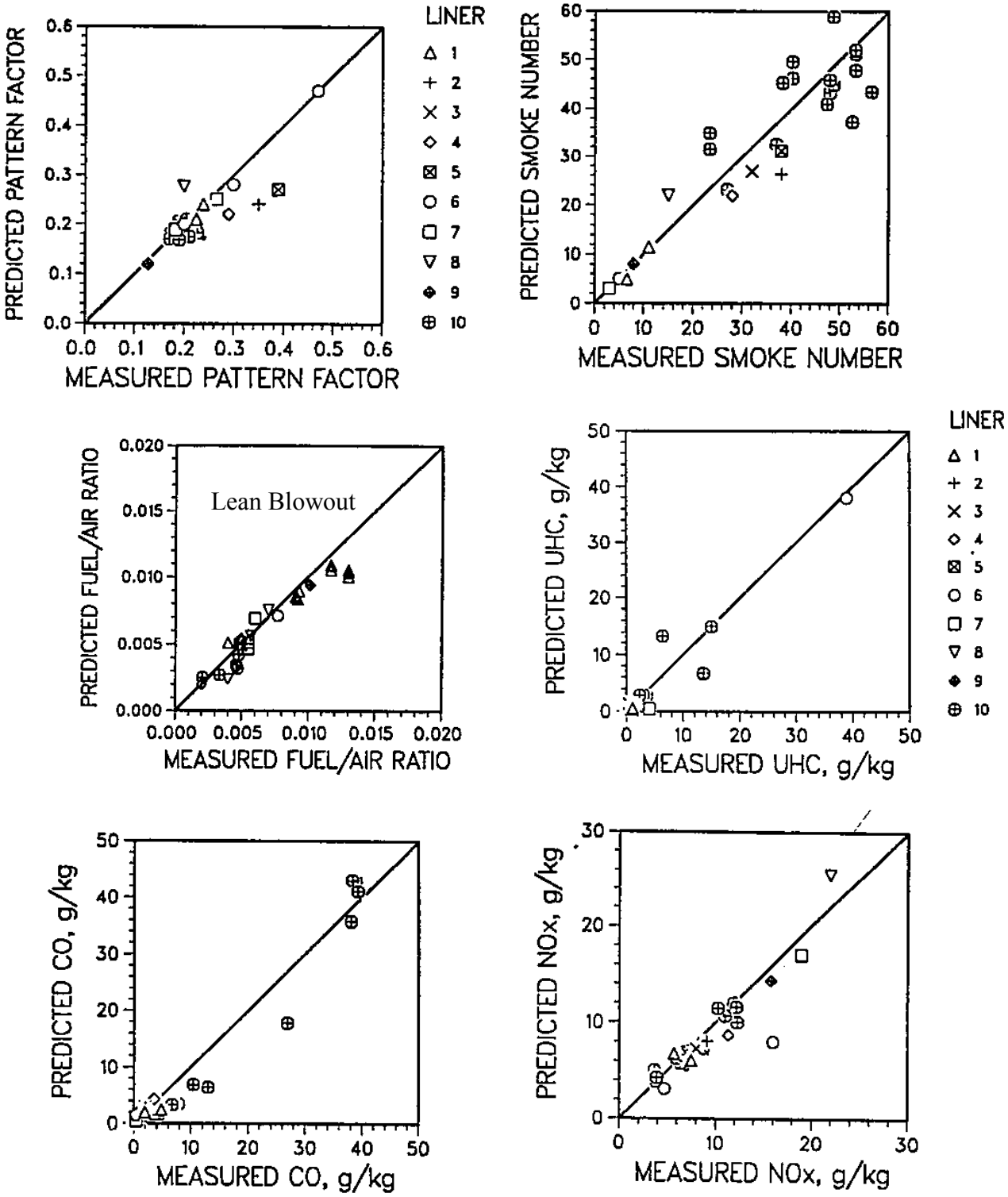


Figure 2: Hybrid modeling gave good agreement w/ 10 combustors data on PF, smoke, lean blowout, HC, CO and NOx.

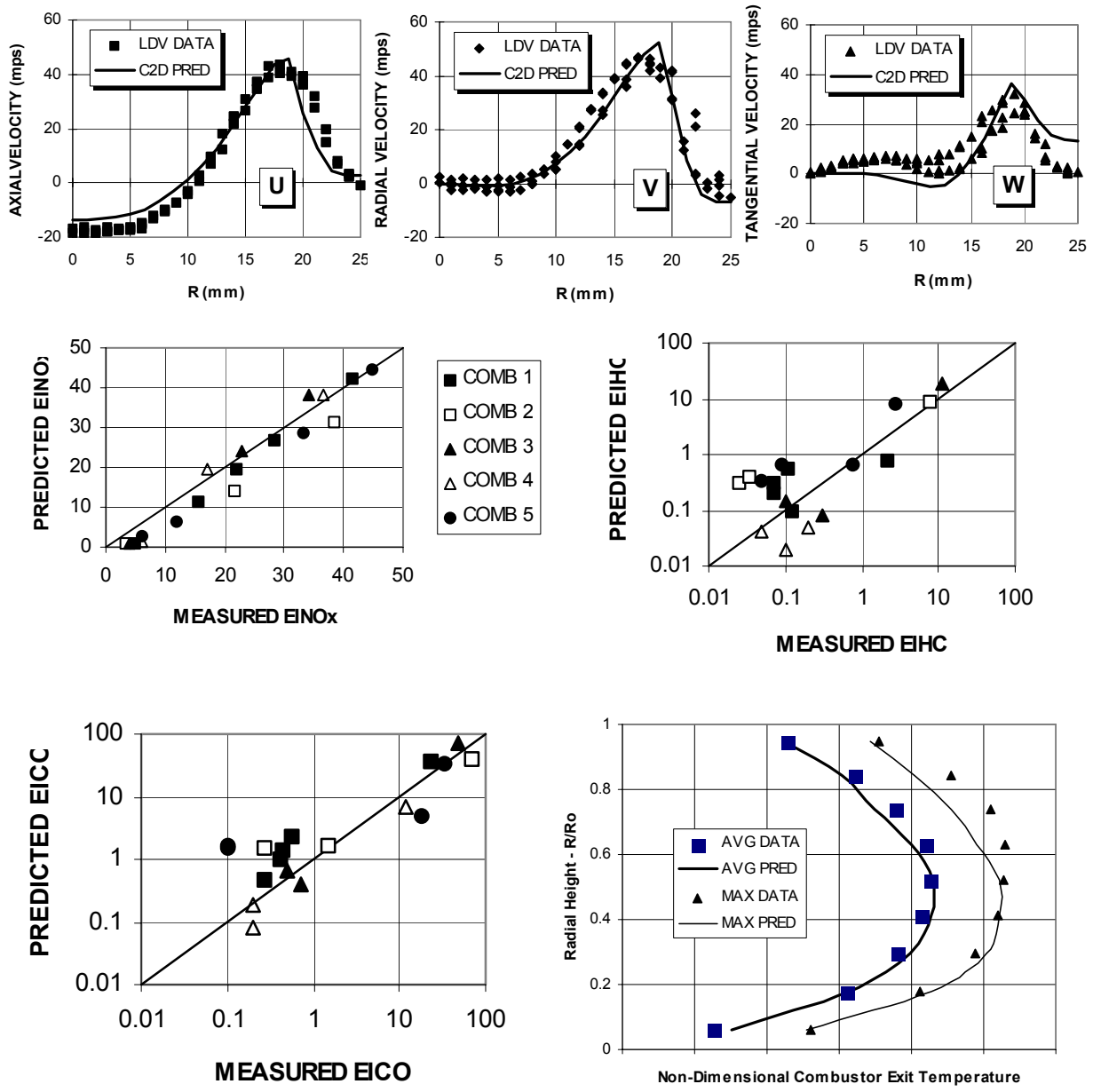


Figure 3: Typical comparison between predicted and measured swirl cup velocity profiles, NOx, HC and CO emissions indices of five propulsion engine combustors, and average and maximum temperature profiles at the burner exit plane. Reported by Mongia (2001).

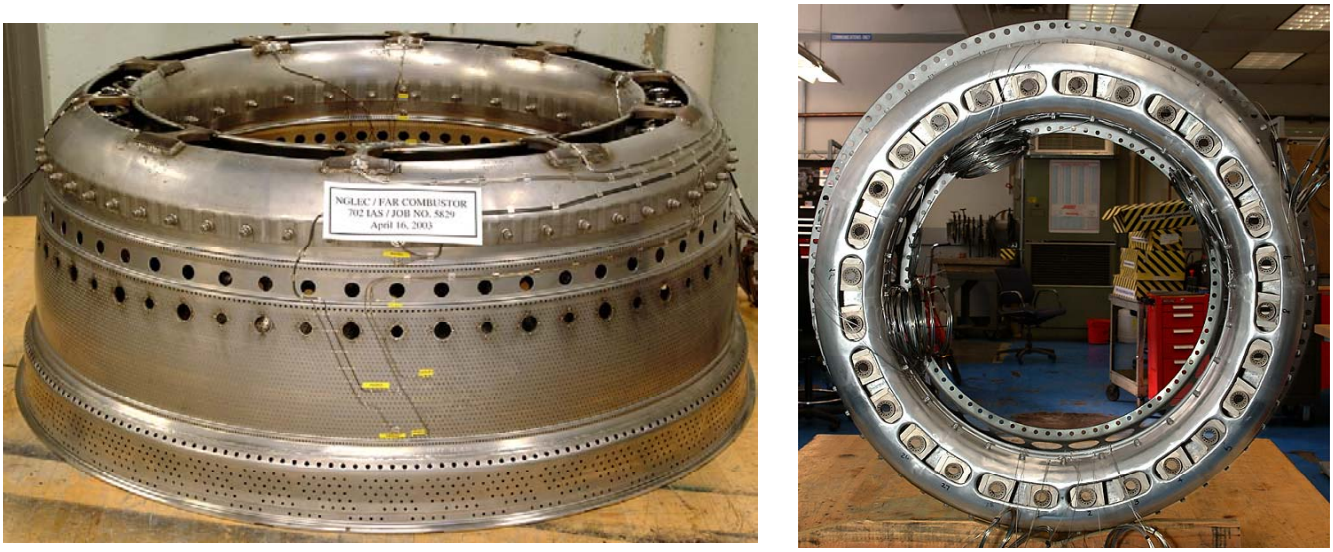


Figure 4: Anchored design methodology has been used in design and development of several SAC and DAC propulsion engine combustors.

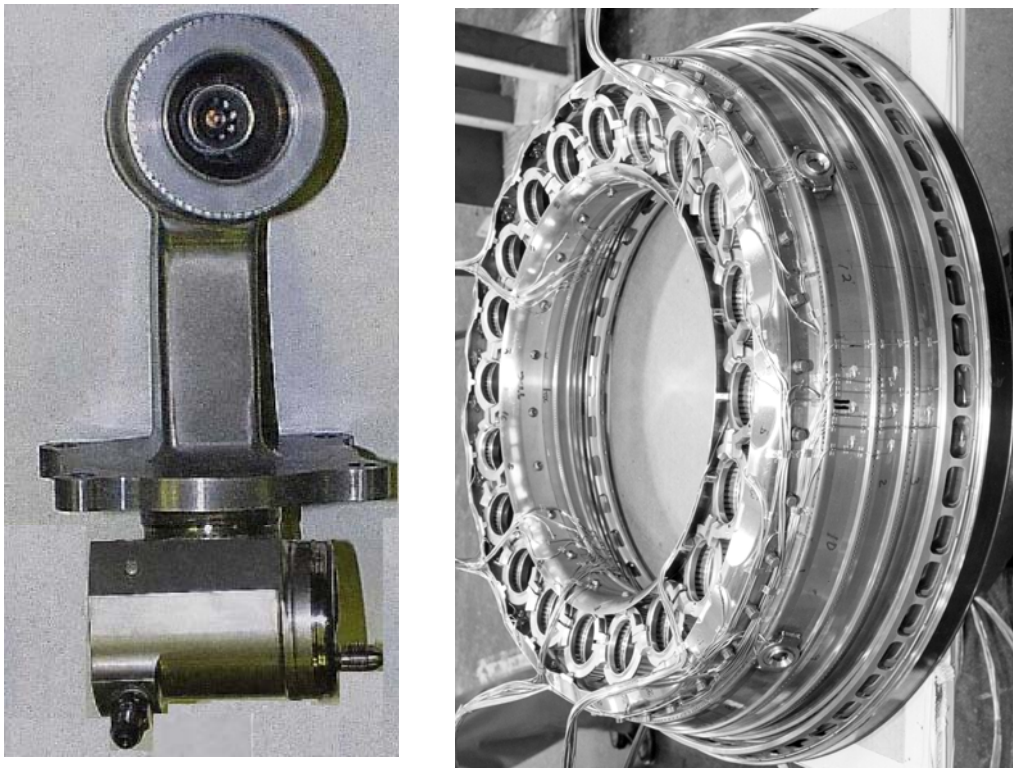


Figure 5: The state-of-the-art CCD-based design tools provided insignificant pre-test insight during the development phase of Twin Annular Premixing Swirlers (TAPS) combustors; Mongia (2003).

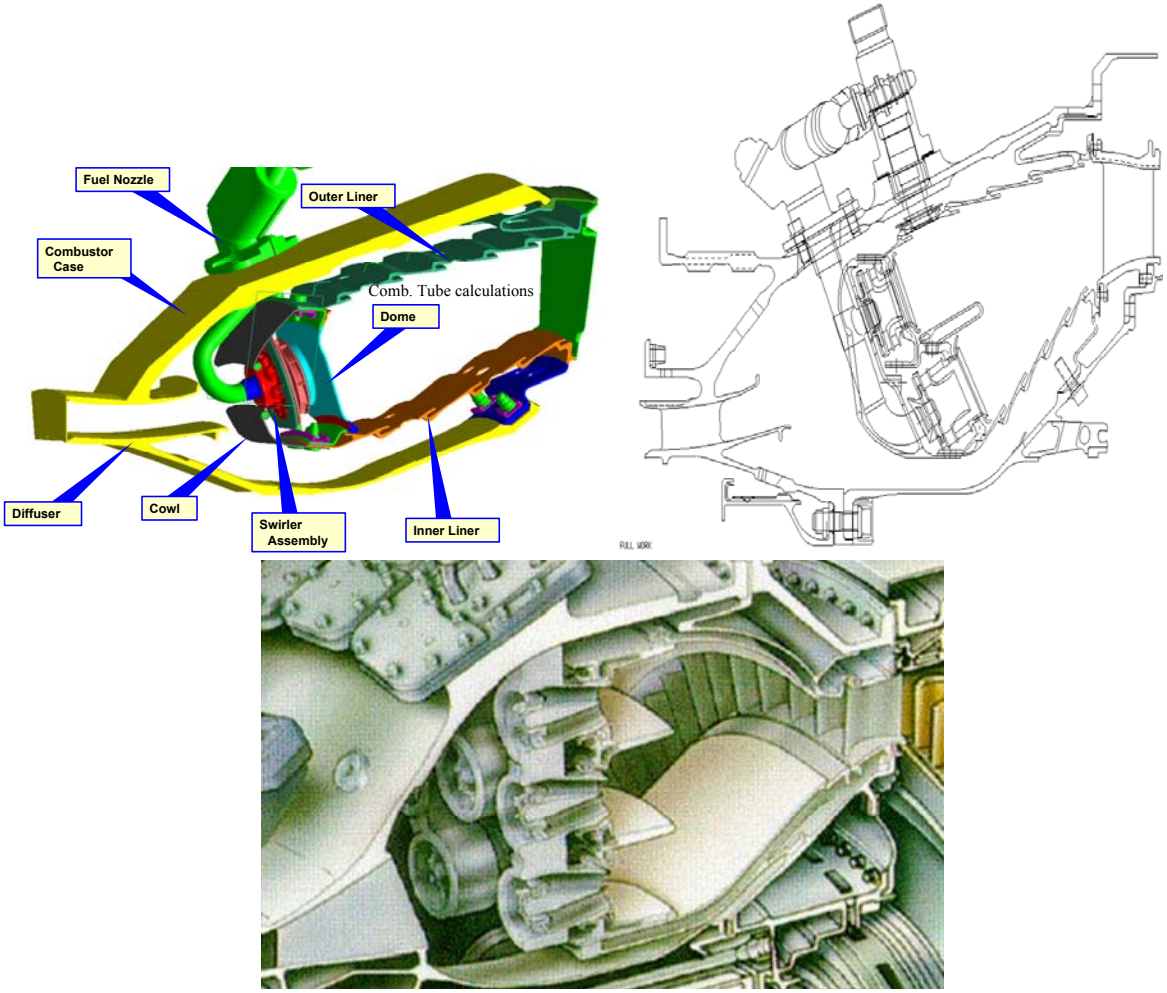


Figure 6: "No-fudge factor" models validation under process for the seven combustors including Single, Dual and Triple Annular Combustors, SAC, DAC and TAC.

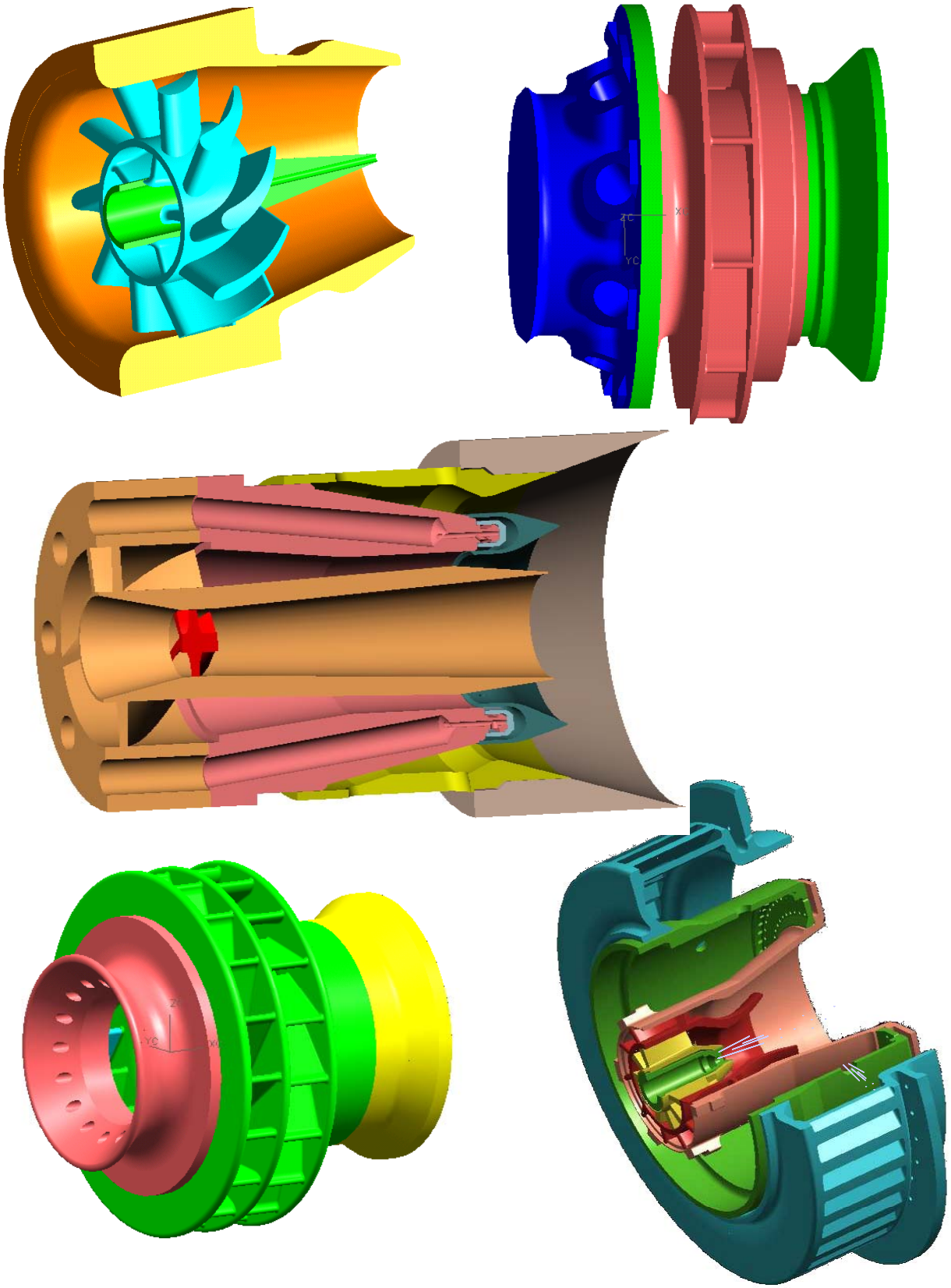
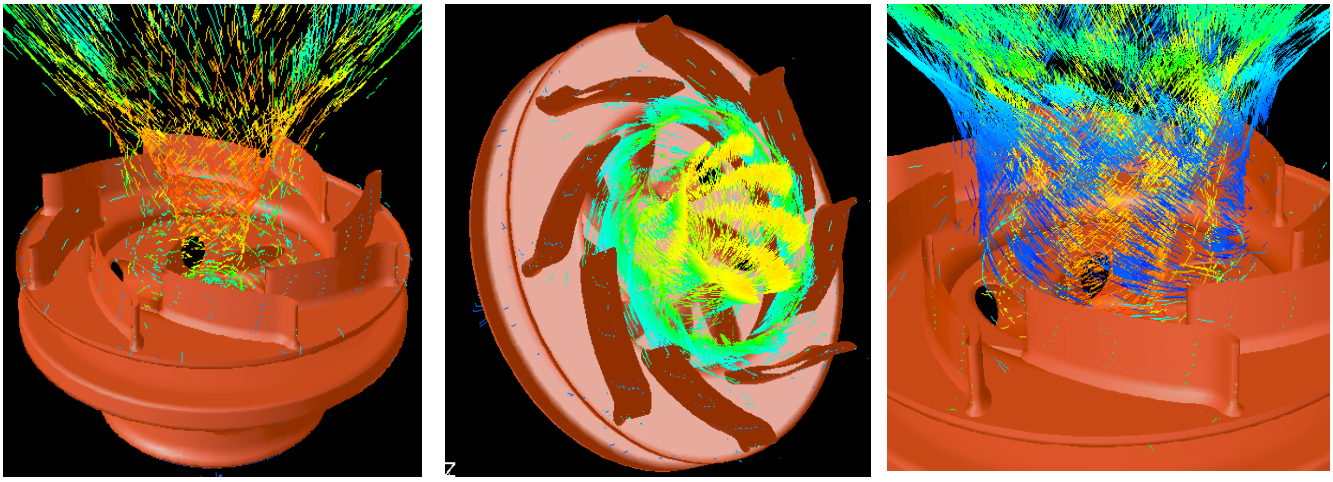
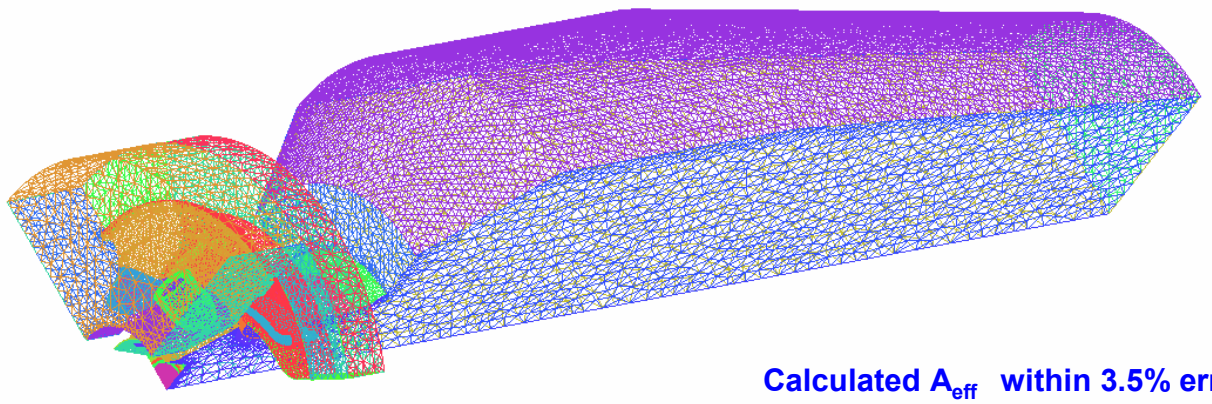


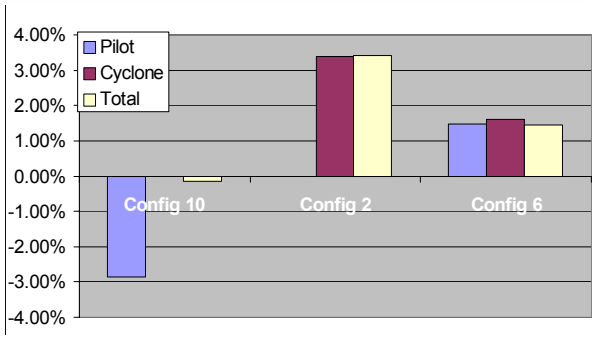
Figure 7: Through the vane calculations have been conducted for five different types of mixers.





Calculated A_{eff} within 3.5% error.

Measured Ae	Predicted Ae	% Error
0.321	0.322	0.19%
0.321	0.323	0.77%
0.321	0.318	0.97%
0.723	0.738	2.07%
0.940	0.953	1.34%
1.000	0.998	0.20%
1.140	1.114	2.28%
1.700	1.710	0.59%
2.320	2.290	1.29%



	Ae (in^2) Nominal	Ae (in^2) Calculation	% Error
Primary	0.132	0.119	9.6
Secondary	0.23	0.236	2.77
Total	0.362	0.355	1.74

Area	Predicted	Measured	Δ
Pilot	0.194	0.2	3%
Main I	0.962	1.0	4%
Main II	1.107	1.14	3%

Figure 8: Through the mixer's inlets (vanes etc.) calculations clearly show complexity of the flow conditions downstream of the vanes. Calculated mixers' effective areas are at least as accurate and reliable as experimental data.

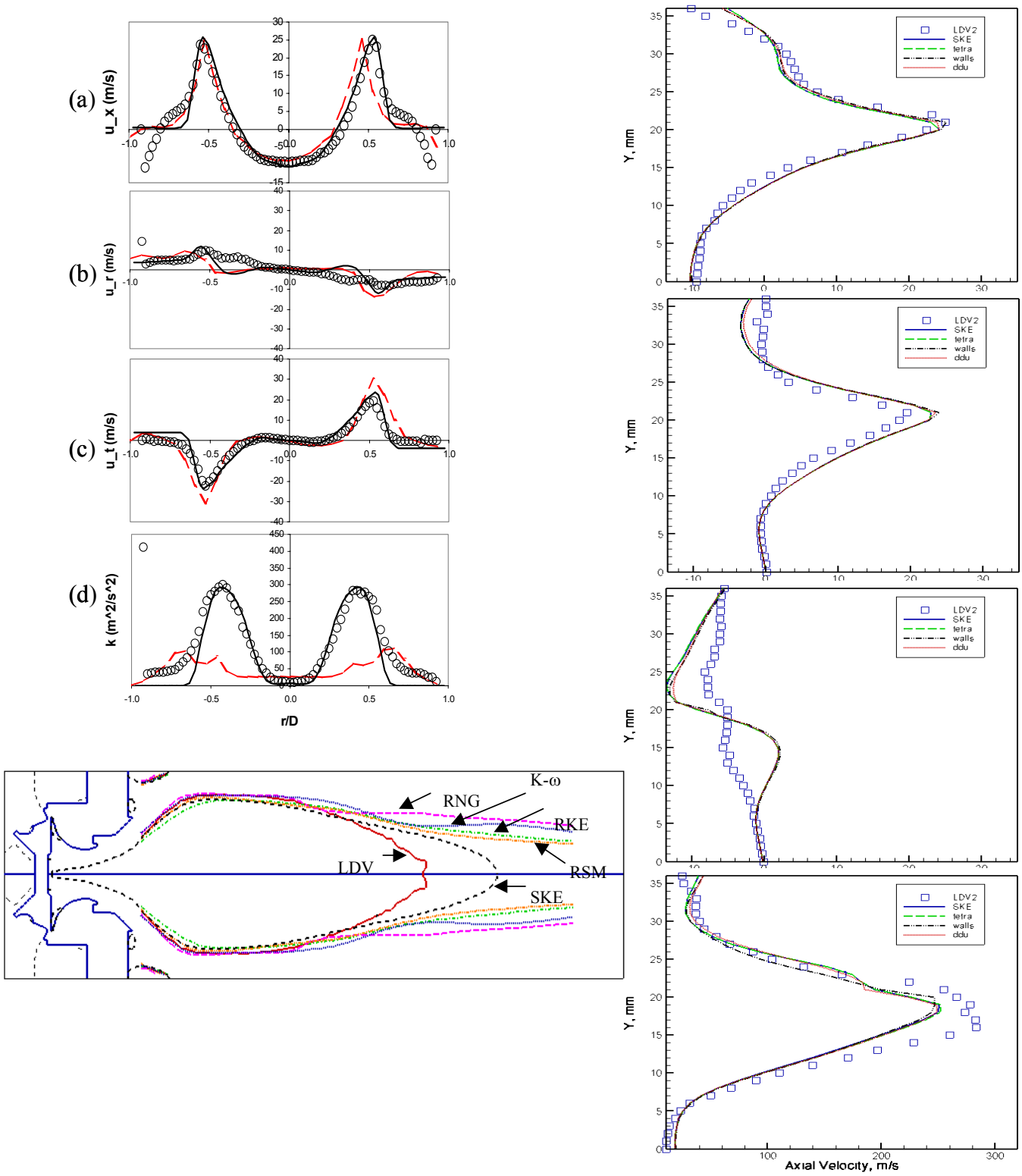


Figure 9: TTV (solid lines) for a swirl cup compared w/ current practice 2-D calculations (dotted lines); symbols denote data. Right hand figures show effect of adaptive gridding on the profiles calculated for X = 3 mm. Recirculation bubble calculated with realizable k-ε, standard k-ε, Renormalized k-ε, k-ω, and the Reynolds stress models of FLUENT; data shown as LDV. From Hsiao et al (2003)

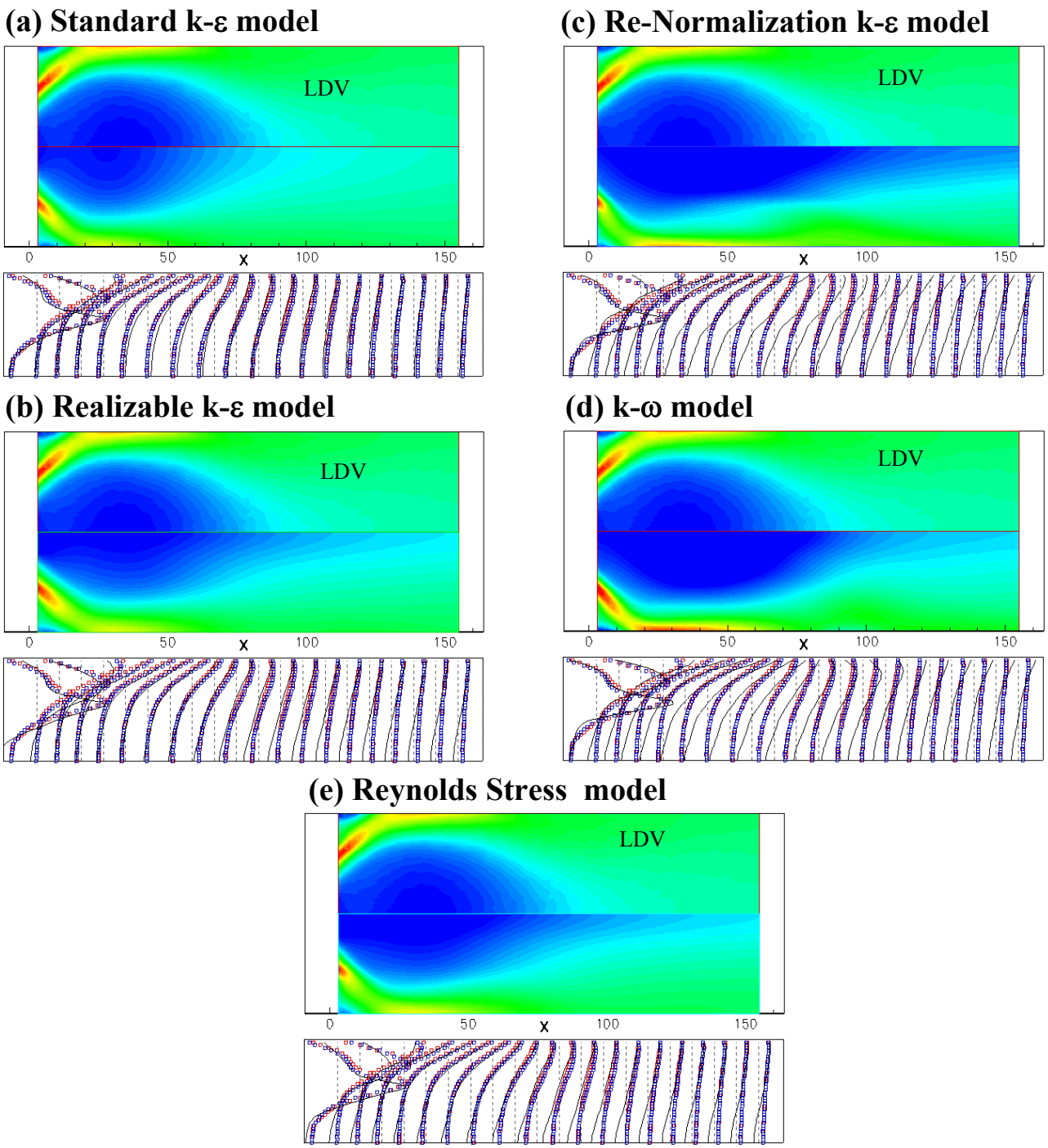
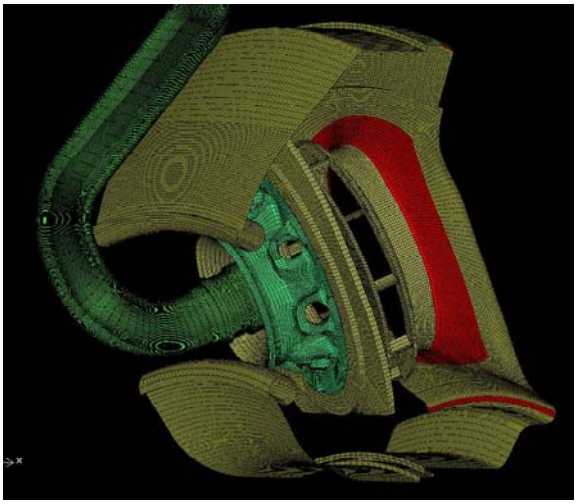
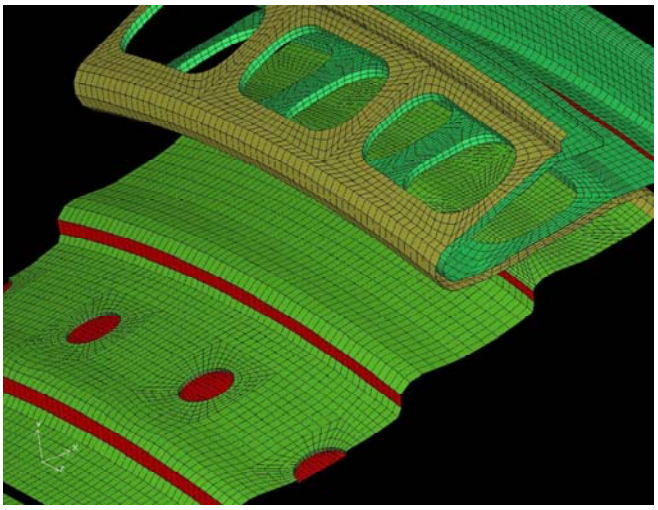


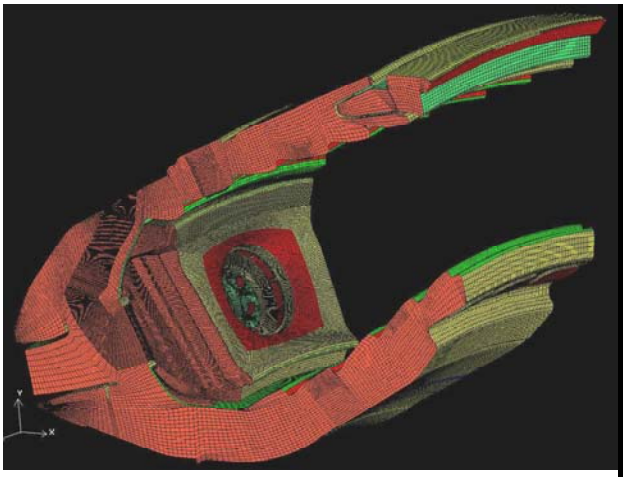
Figure 10: Standard k-ε model might not be as bad as thought over the last 20 years for application to swirl cups. Axial velocity contours downstream of a swirl cup calculated with realizable k-ε, standard k-ε, Renormalized k-ε, k-ω, and the Reynolds stress models of FLUENT; data shown as symbols or LDV. From Hsiao et al (2003)



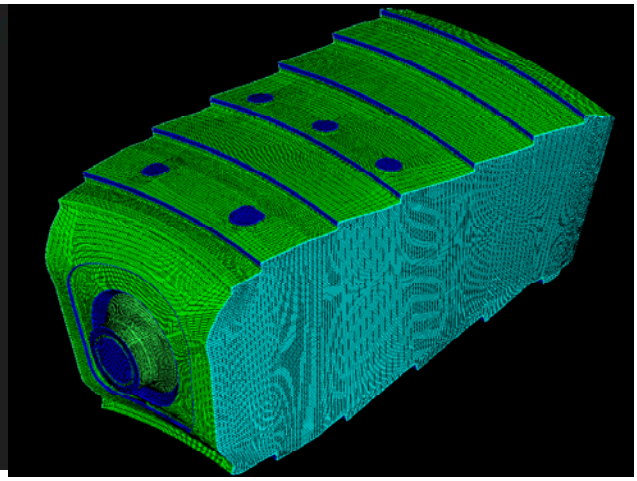
(a) Swirl cup, dome & part of nozzle



(b) Part of liner and dilution holes



(c) Diffusion system including swirl cup - 2.2 M nodes



(d) Combustor w/ one million nodes

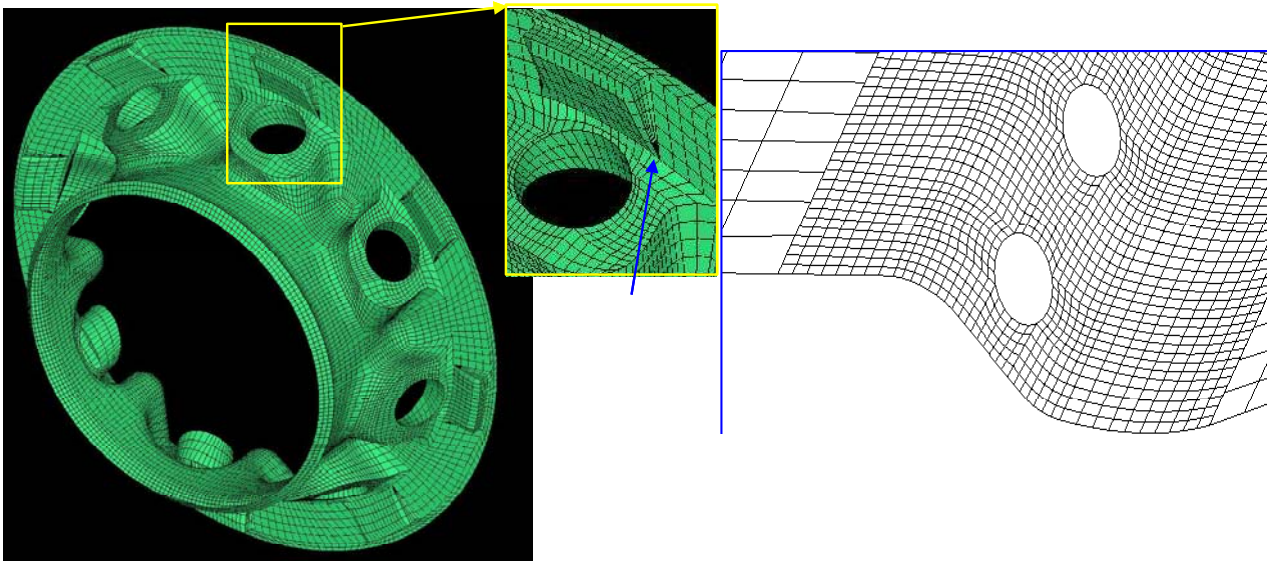
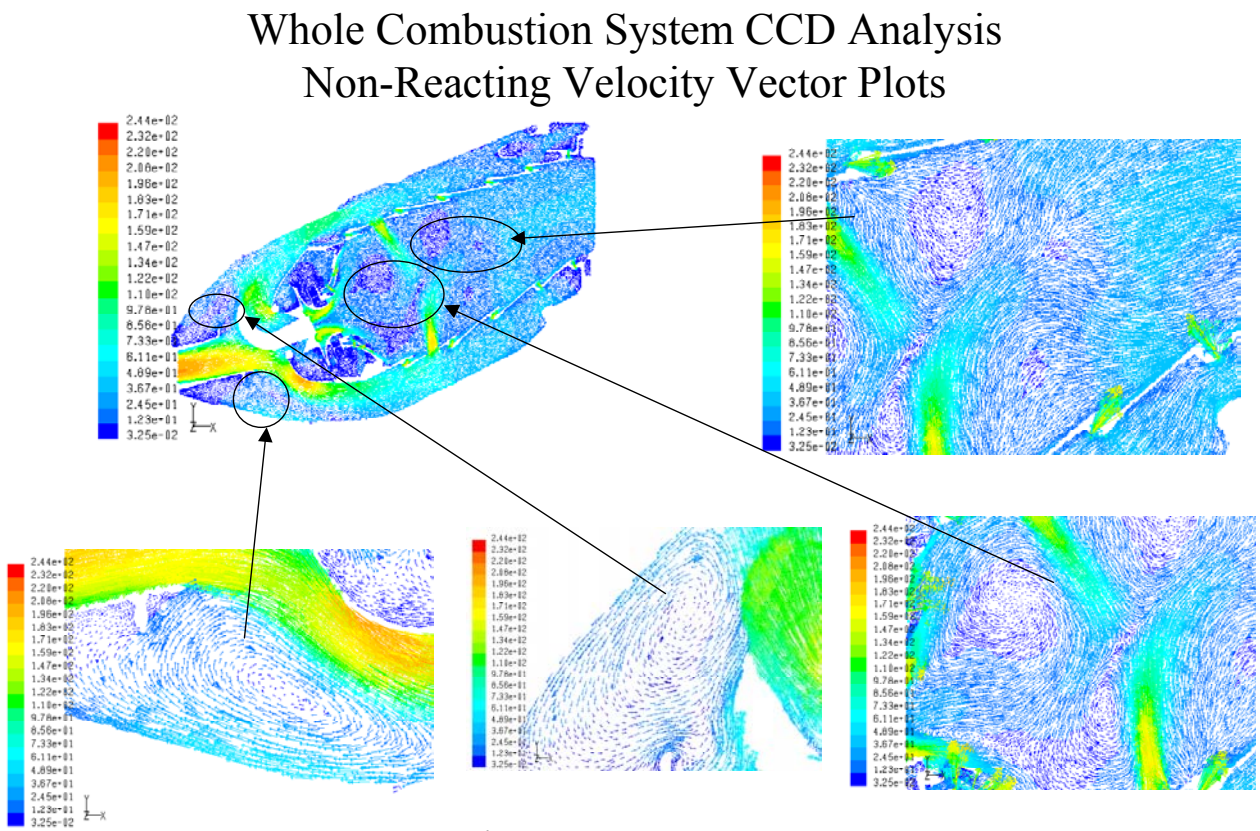


Figure 11: Grid generation technology (time-consuming as it is) in place for resolving finer details of combustion systems required for accurate prediction of aero-thermal field.



Velocity vectors

Figure 12: Integrated combustor/diffuser calculations may be of "show and tell" value unless we address issues about:

1. Pressure and airflow distribution around the combustor
2. Hydrodynamic validation of combustor liner subcomponents including mixer (fairly under control), primary/dilution holes, splash plate and cooling including nuggets and multi-hole.
3. Cold- and hot-side convective heat transfer.
4. Comprehensive injector/mixer modeling.
5. Turbulence/chemistry interaction including skeletal hydrocarbon and detailed Nox chemistry.
6. Soot formation/oxidation and radiation modeling.

Table 2: Preliminary assessment for predicting pressure recoveries of a typical diffuser set-up.

	Center-peak Profile				Inner-peak Profile			
	Data	A	B	C	Data	A	B	C
Prediffuser exit CP	0.496	0.603	0.485	0.520	0.505	0.604	0.494	0.516
Outer Passage CP	0.504	0.654	0.563	0.532	0.517	0.655	0.562	0.531
Inner Passage CP	0.534	0.672	0.576	0.554	0.530	0.672	0.587	0.557
Mean square root error, %		13.30%	4.26%	2.43%		12.76%	4.20%	1.84%

A: 2-D CFD w/ measured diffuser inlet profiles
B: 2-D CFD w/ through-the-inlet-profiler calculation for inlet conditions
C: 3-D CFD w/ through-the-inlet-profiler calculation for inlet conditions

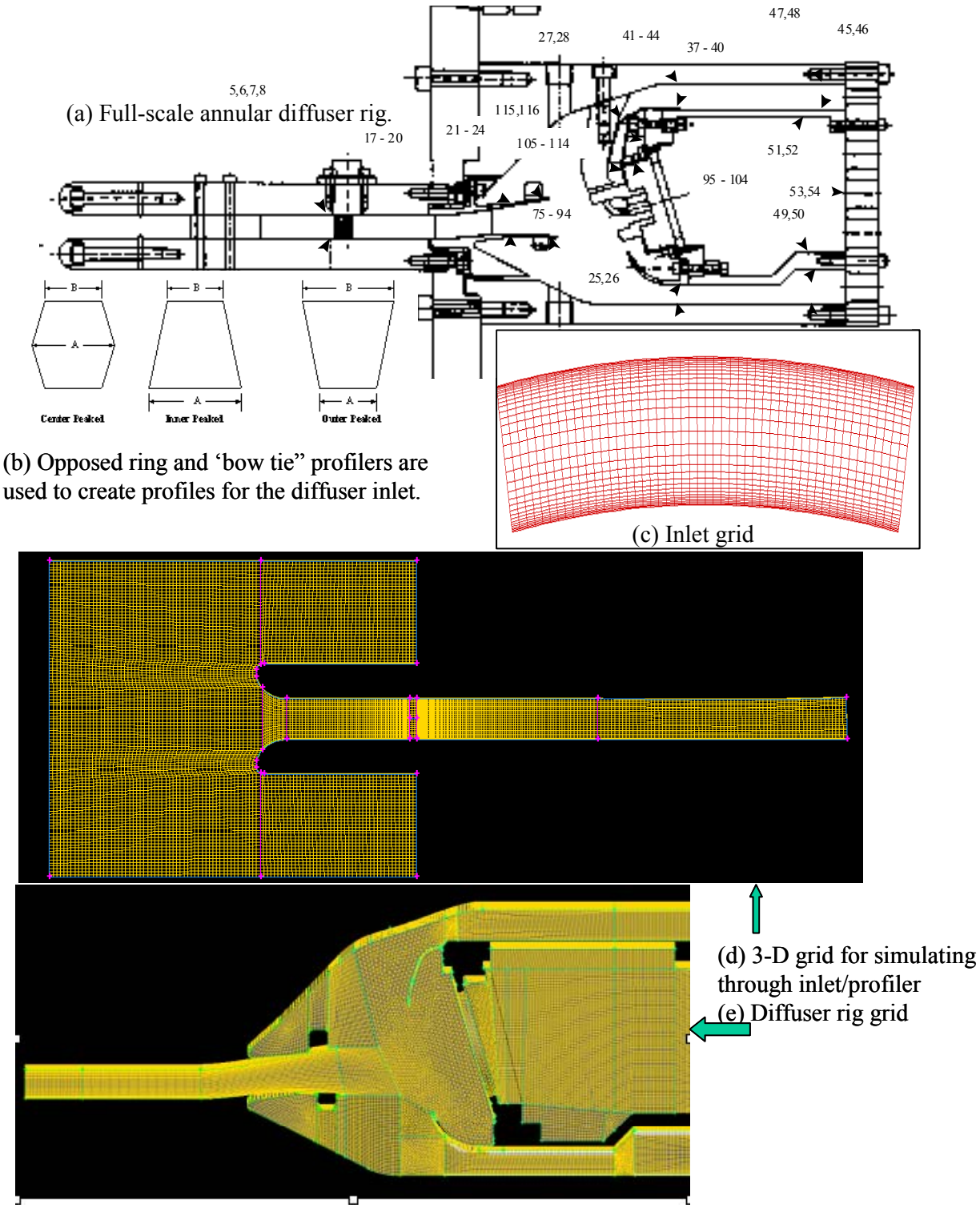
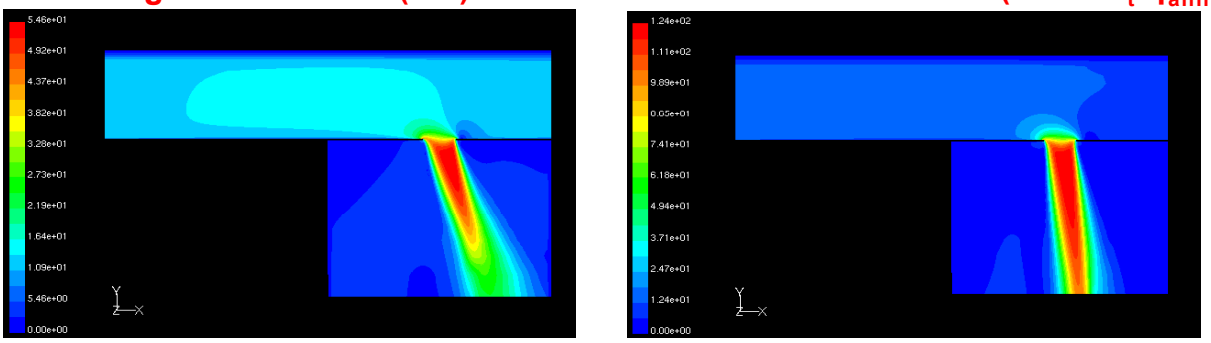


Figure 13: Diffuser rig schematic and typical grid used for inlet and diffuser calculations.

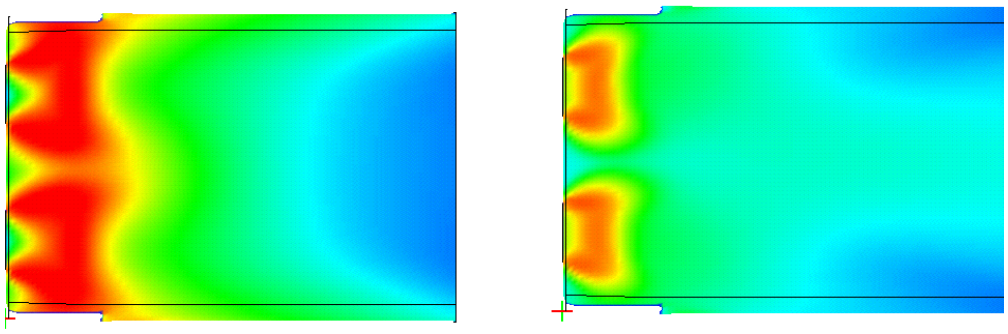
Discharge Coefficient (Cd) Vs Pressure Ratio Parameter ($K = \Delta P_t/q_{ann}$)



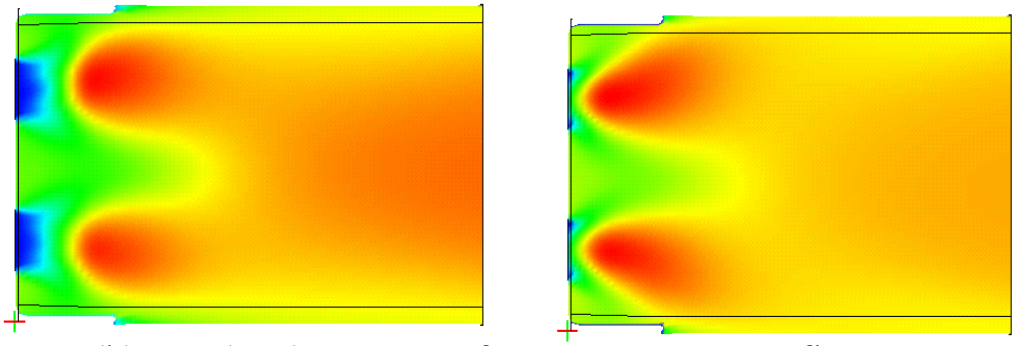
K =10 **K =100**

K	Cd_CFD	Cd_Data	% Error	Jet angle
10	0.5624	0.598	-5.95	64.92
20	0.590	0.599	-1.50	69.95
50	0.601	0.603	-0.37	76.82
100	0.612	0.610	0.33	80.36

Figure 14: Through the primary/dilution holes calculations show reasonable agreement with Cd and jet angle data from Giri et al (2003).



(a) Log(CO) contours for 1750 & 1900K flames



(b) Log(NO) contours for 1750 & 1900K flames

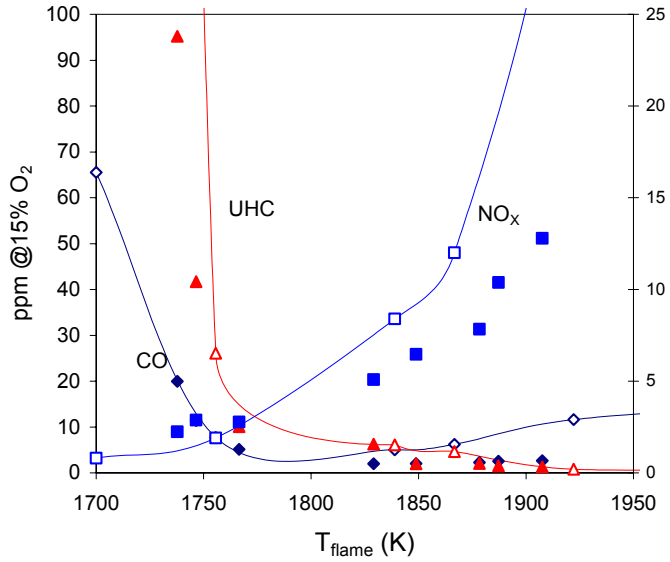


Figure 15: Laminar flamelet calculated by Held et al (2001) NOx, CO and HC for a Dry Low Emissions 2-cup compare well with data, with assumed profiles at the exit of DACRS mixer, i.e., without going through TTV calculation process.

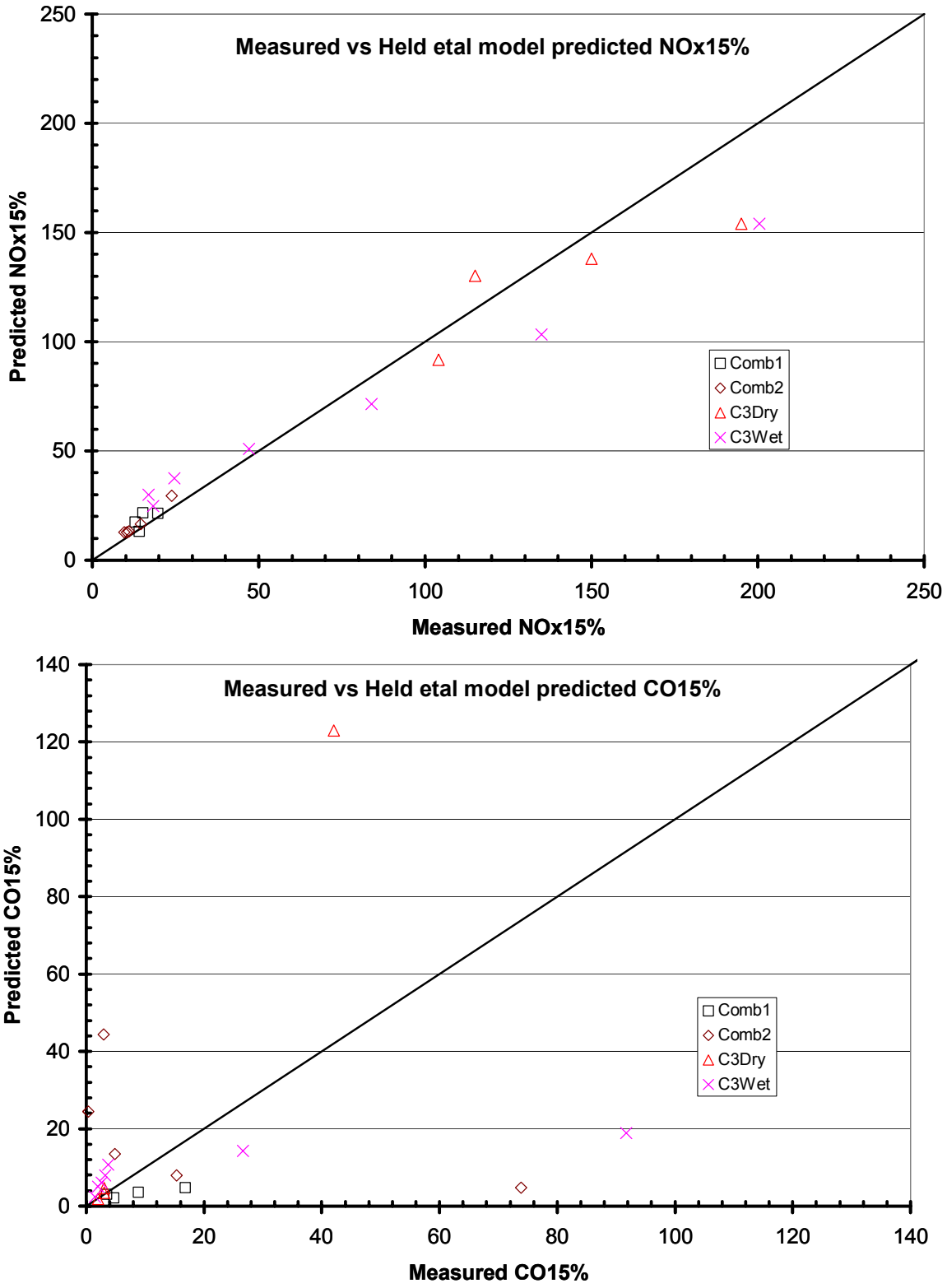


Figure 16: Comparison between measured and predicted (w/ Held et laminar flamelet model, LFM) Nox and CO data of three additional combustors identified as Comb2, C3 (dry and wet) and C4.

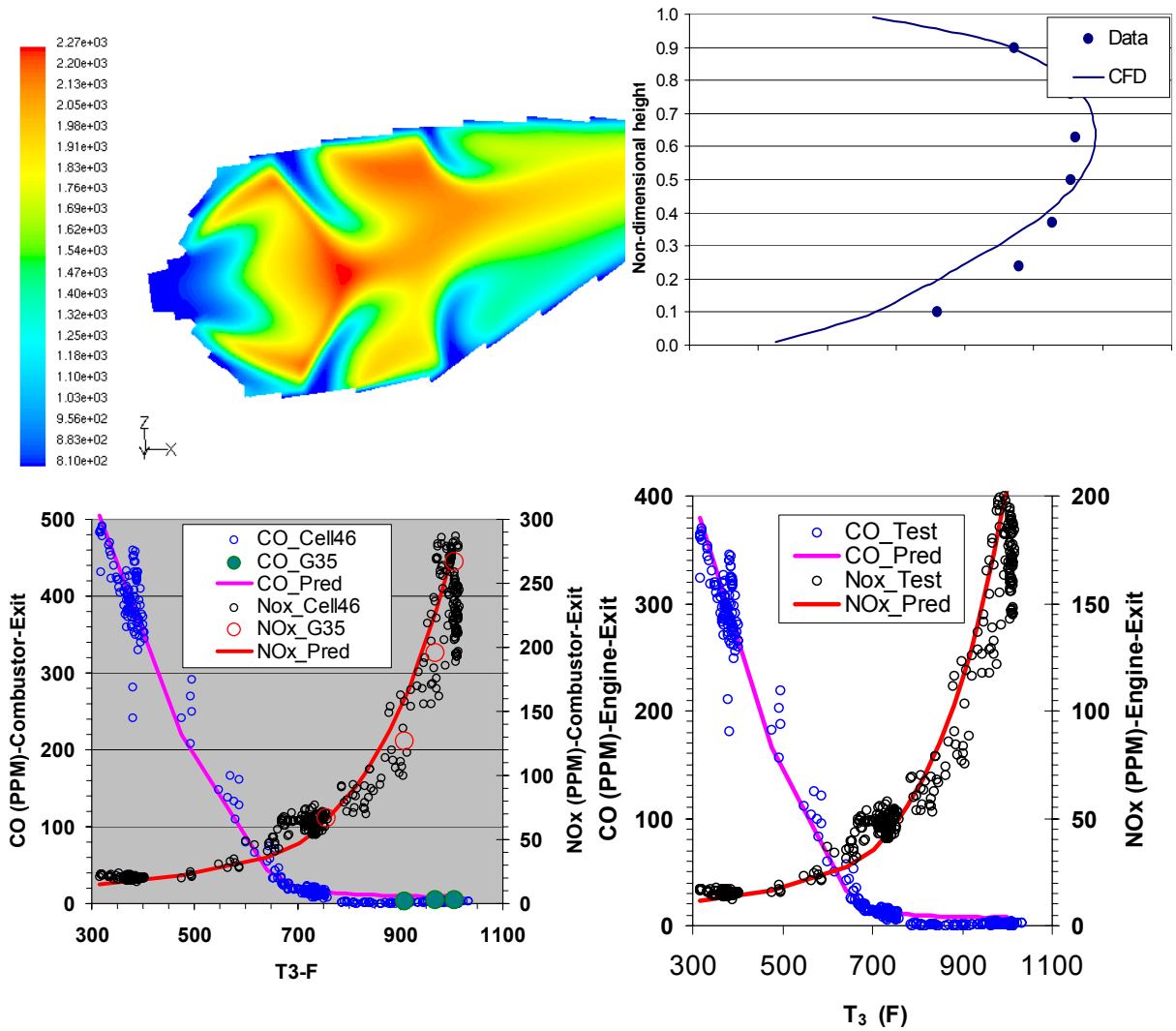
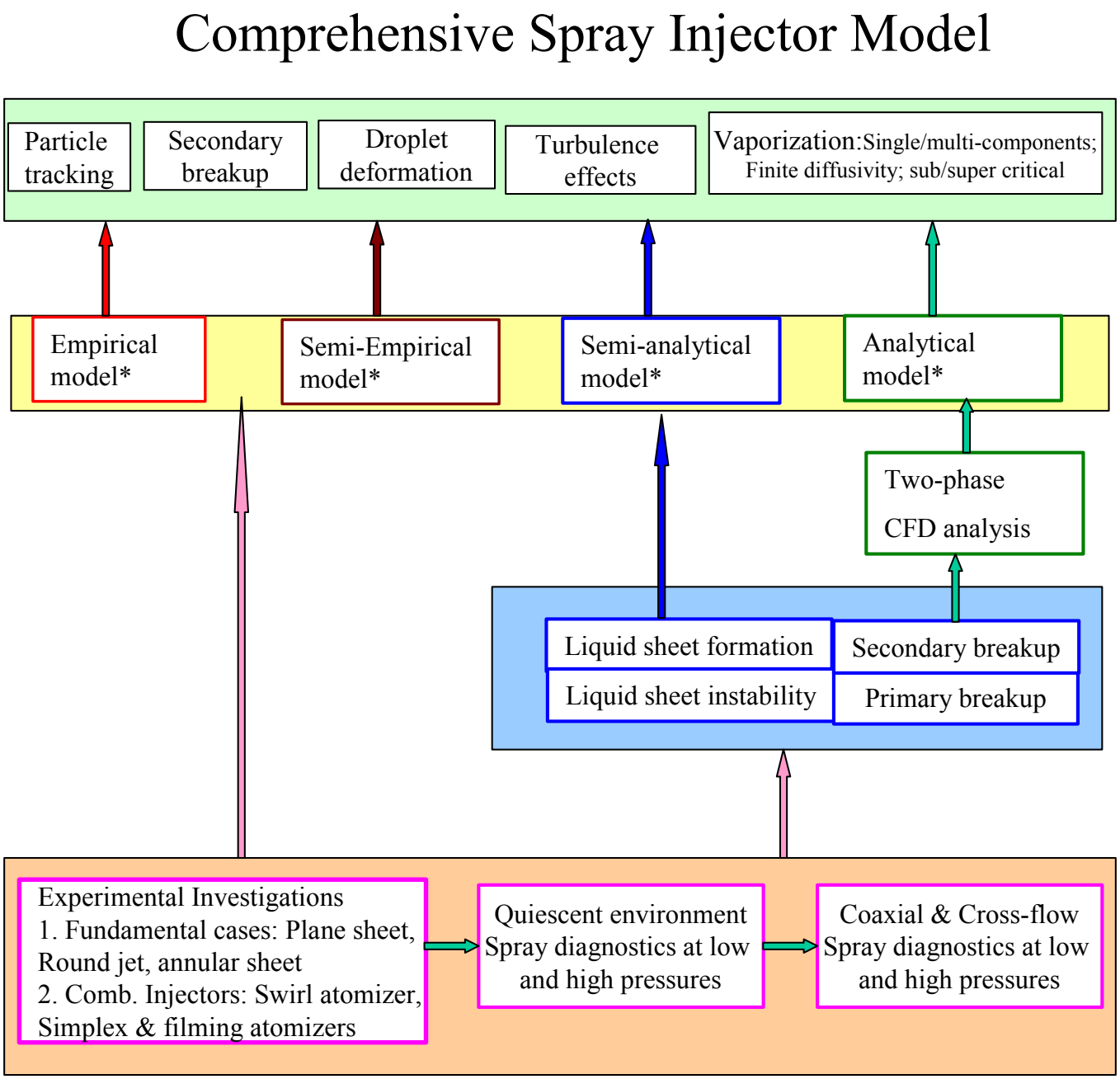


Figure 17: Comparison between data and calculated (through the inlet calculation approach) NOx and CO of a NG fired rich-dome swirl cup combustor with a laminar flamelet model of Li et al (2003).



* For droplet size distribution, mean and rms velocity components.

Figure 18: Comprehensive model development plan.

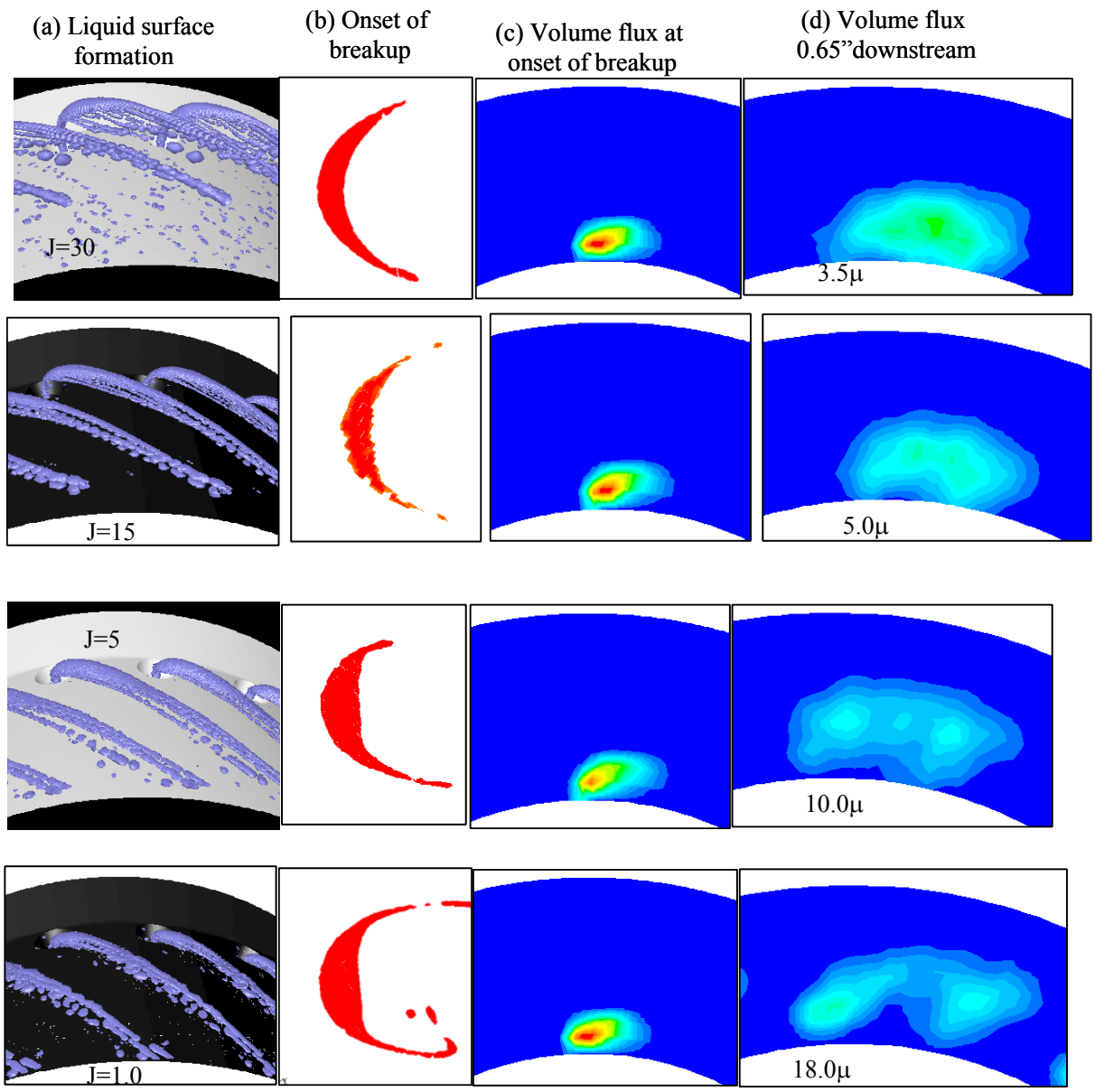


Figure 19a: Comprehensive mixer analysis- typical results for a mixer involving cross-jet atomization.

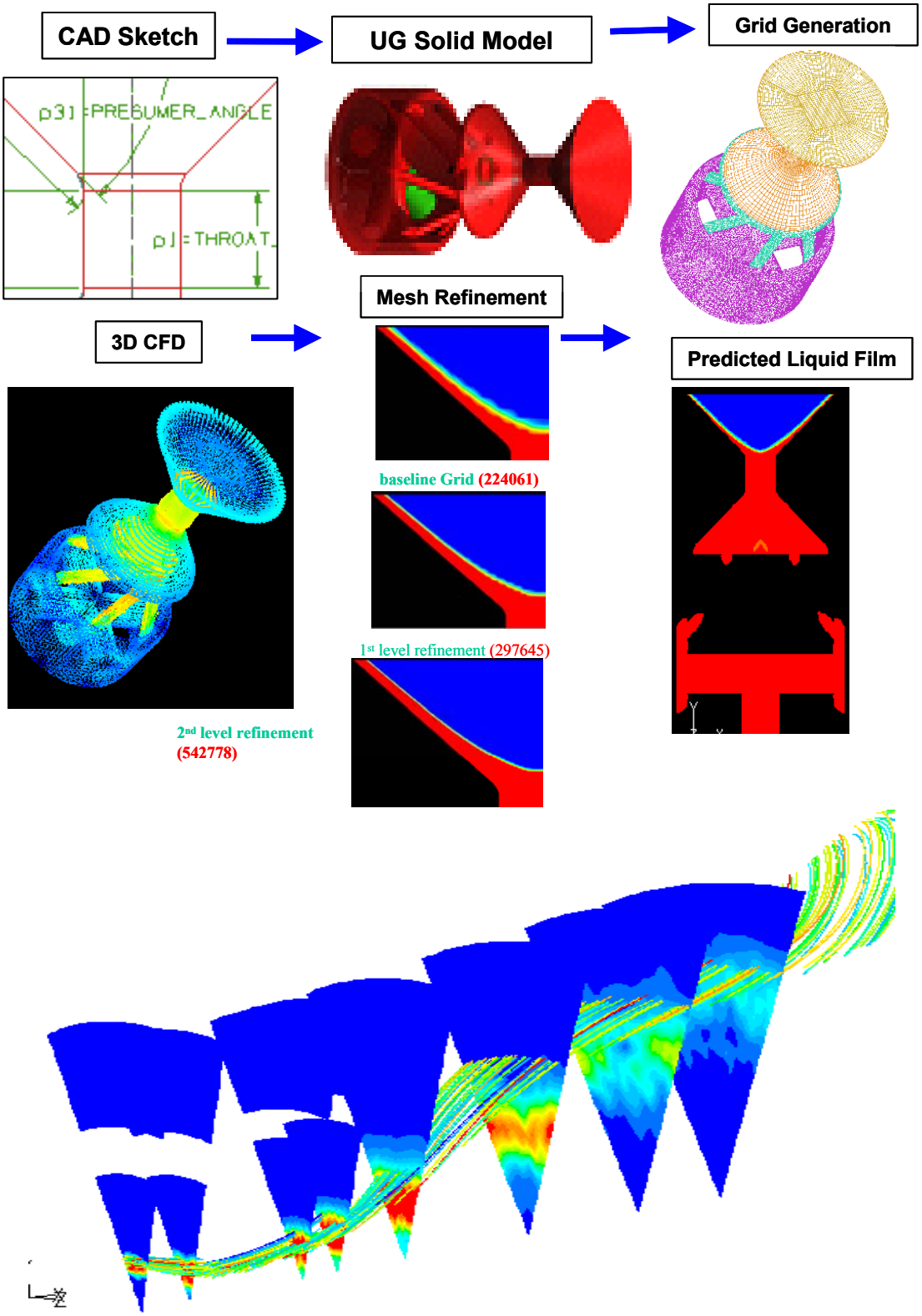
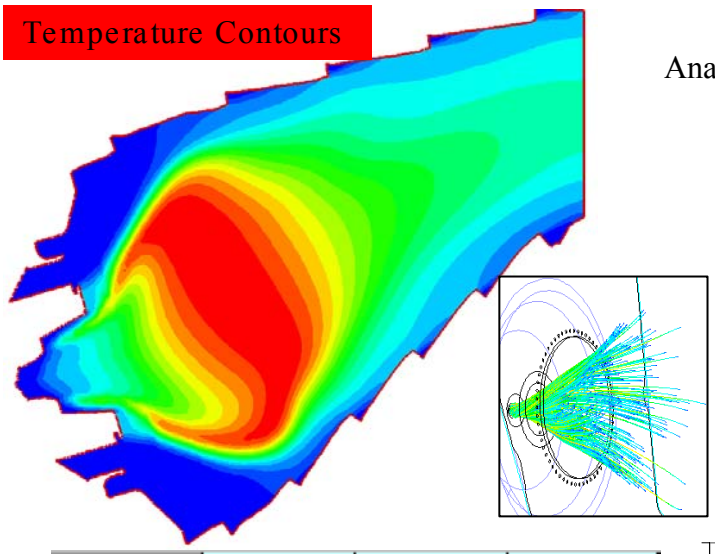


Figure 19b: Comprehensive mixer analysis- typical results for a mixer involving simplex pressure atomization followed by air atomization.



Analytically predicted spray characteristics

Low-Power Operation	D32	Min Dia	Max Dia
# 1	75.1	12.4	115.1
# 2	39.9	6.9	72.3
# 3	18.6	3.1	28.6
# 4	20.6	3.6	37.4

	CO ppm	NO ppm	HC ppm
DATA	1147	29	21
CFD	915	32	24
DATA	521	35	3.42
CFD	447	37	0.33
DATA	99	87	0.4
CFD	75	97	0.001
DATA	56	124	0.31
CFD	58	146	0.002

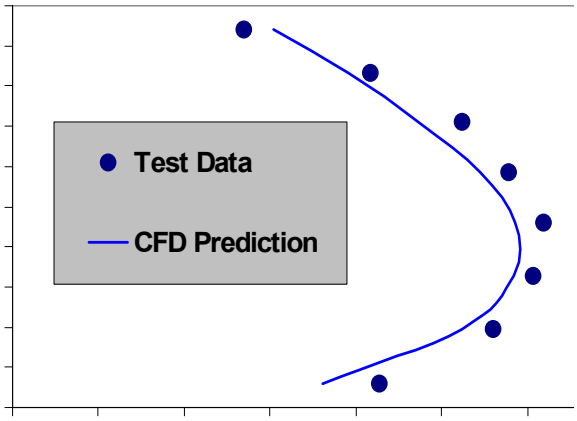


Figure 20: Typical results for temperature field, spray quality at low-power and comparison between predicted (w/ Li et al Laminar Flamelet Model, LFM) and measured combustor exit emissions and profile of a Single Annular Combustor.

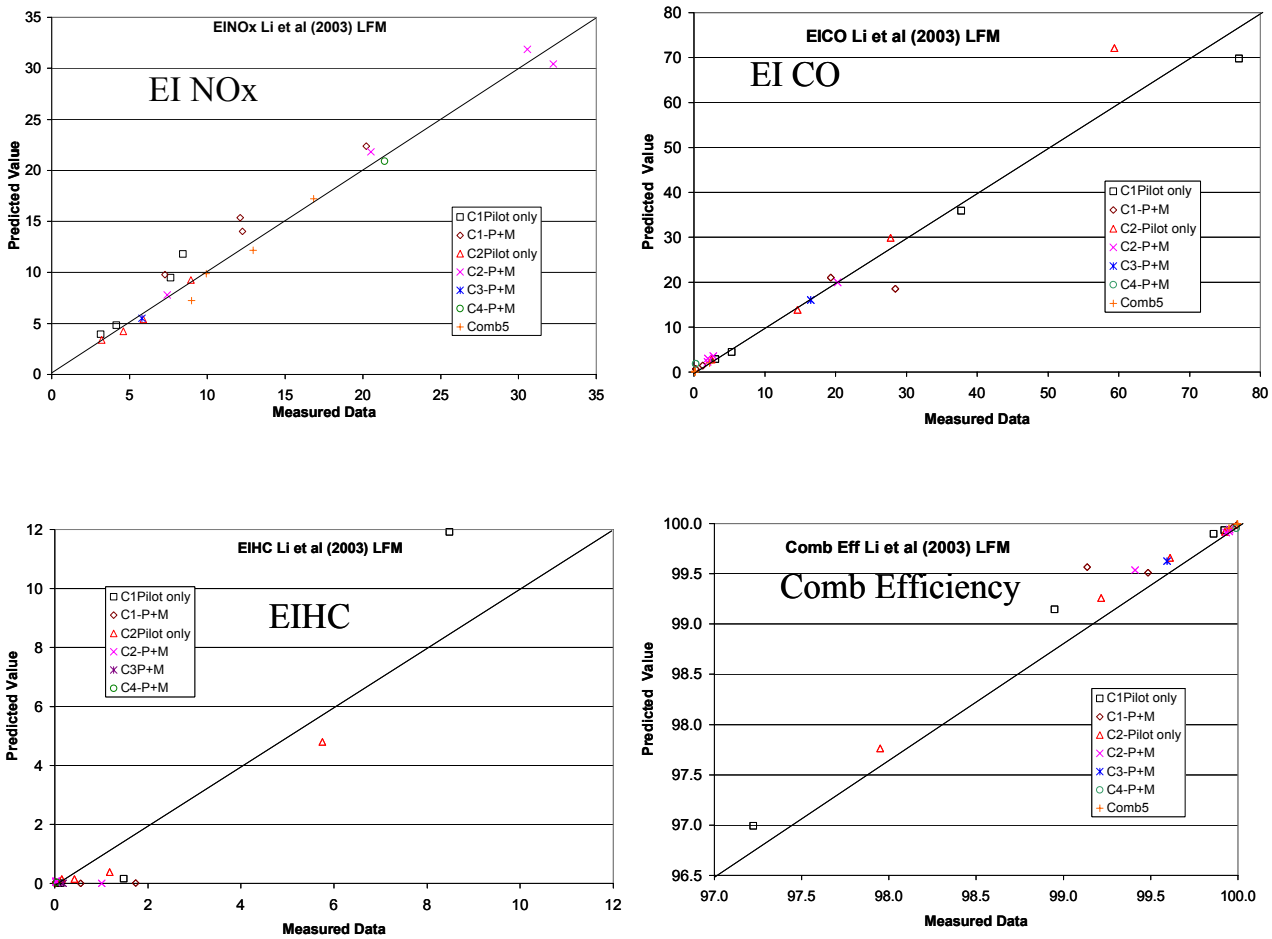
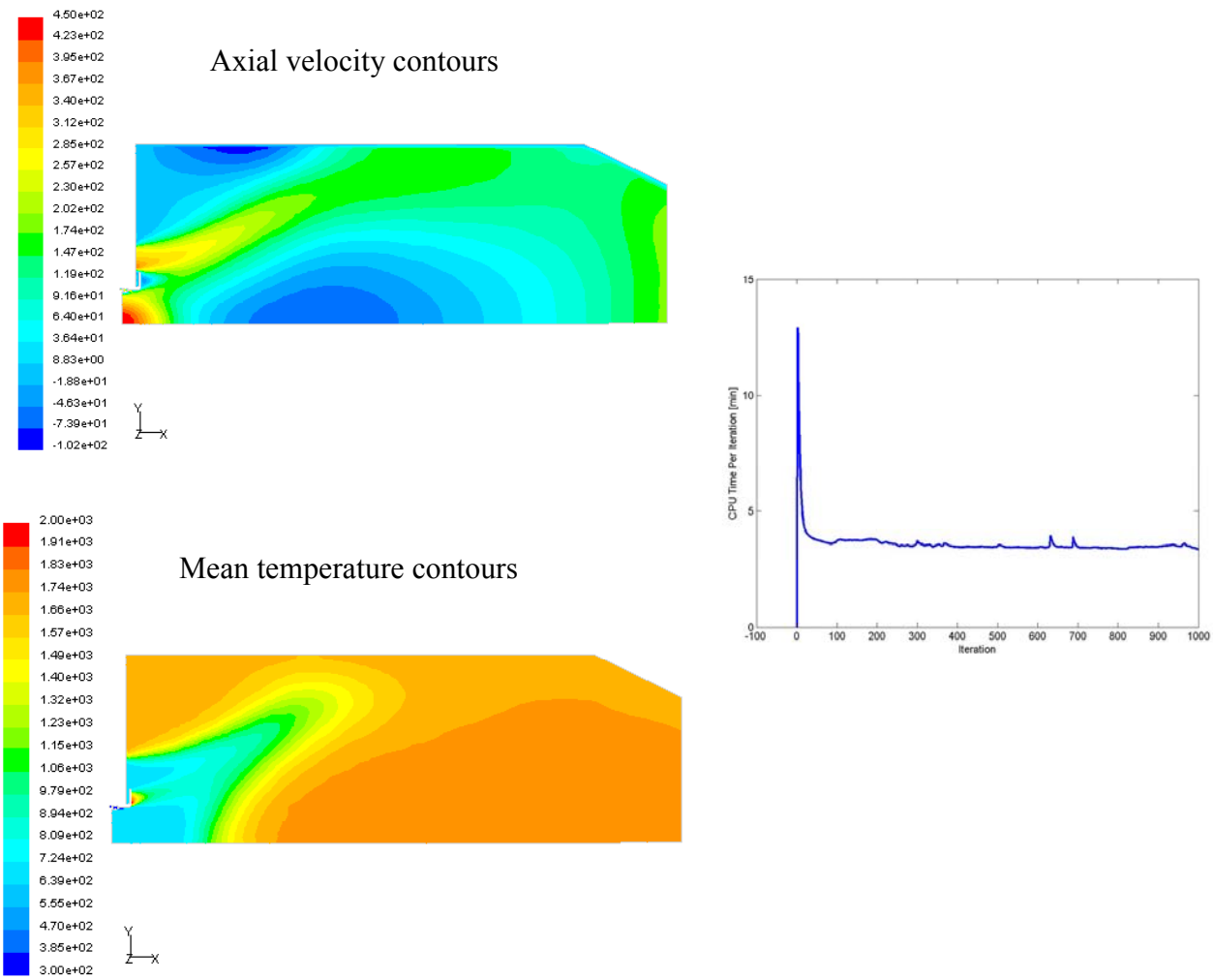


Figure 21 Comparison between measured and predicted (w/ Li et al Laminar Flamelet Model, LFM) Nox, CO, HC and combustion efficiency of the five gas turbine combustors.



Case	Bulk equilibrium temperature based on total F/A [°F]	Pilot equilibrium temperature based on F/A [°F]	NO (corrected to 15%O2)		CO (corrected to 15%O2)	
			Data [ppm]	CFD [ppm]	Data [ppm]	CFD [ppm]
1	2700	3100	8.9	2.7	70	21
2	2900	3100	10.5	4.1	5	9.1

Figure 22: Joint scalar PDF approach and perspectives: PDF vs laminar flamelet .